

W. M. KECK OBSERVATORY ADAPTIVE OPTICS NOTE
KAON 1556

The Keck AO Performance Estimator GUI

A user's manual for the interactive interface

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Table of Contents

1. What this is, and what it is not.....	4
2. Getting started.....	4
3. The input tabs.....	4
4. Entering an offset three ways.....	13
5. Reading the plots.....	13
6. Measuring real frames: the Measured SR tab.....	17
7. What-if, reports, and honest provenance.....	20
8. Predicting performance for conditions you choose.....	21
9. Saving your work.....	25
10. If a number looks wrong.....	25

List of Figures

Figure 1 — the main window after a Run: input tabs at left, the Timeline plot at right, and t.....	4
Figure 2 — Nighttime mode enabled: the Data-tab checkbox, the source-selection controls greye.....	5
Figure 3 — the summary-stats panel with STRAP selected: NGS, single-beacon-LGS, and LTAO Stre....	6
Figure 4 — the same panel with TRICK selected: every K2 cell reads n/a, since a K1-only senso.....	6
Figure 5 — the FA advisory on the Data tab, triggered: the “⚠ FA event in progress” line with.....	7
Figure 6 — the FA geometry dialog for the M79 frames of 2026-01-31: plan view of the monitor	7
Figure 7 — the starlist picker with a row selected: the detail panel, the HA column, and the	9
Figure 8 — (a) the Load starlist... picker with three TT-star candidates linked to one target;	9
Figure 9 — the Target tab: a saved target list, the Target offset control set to a nonzero sh.....	10
Figure 10 — the NGS tab: guide-star magnitudes, the three-way offset entry, and the editable G.....	11
Figure 11 — the LGS tab with TRICK (K) selected as the tip-tilt sensor and the night’s real Cn.....	12
Figure 12 — the WFE sliders tab, with the Budget values picker at 3_1_3. Each slider sets a bu.....	12
Figure 13 — the TT-star offset (LGS tab) with “fix to base position” ticked, entered as a tota.....	13
Figure 14 — the field-of-regard view with a DSS2 sky backdrop: the 60” patrol circle, the scie.....	14
Figure 15 — the right-click menu with both actions visible — dropping a target and redefining	15
Figure 16 — the field map (K1 single-LGS, science field) with three science-target markers dro.....	15
Figure 17 — the catalogue overlay with the current unfilled-ring, size-graded markers (GSC 2.4.....	16
Figure 18 — the same view at Field PA = 0° (left) and 60° (right): the backdrop, outlines and	16
Figure 19 — a loaded frame unflipped (left) and with Frame flip: X ticked (right).....	17
Figure 20 — the popped-out log on its Image log tab, with the Guide mag source column showing	19
Figure 21 — the Measured SR tab mid-pick on an M79 OSIRIS frame: the PICK ZOOM magnifier locke 19	
Figure 22 — the measured field map on the same frame with EE correction on: 30 quality stars k.....	20
Figure 23 — Report = FWHM on the Timeline, showing the two-panel FWHM view. The FWHM conventio21	
Figure 24 — the Prediction scenario in a fresh GUI with no night loaded: the field map render.....	22
Figure 25 — the Error terms tab under the same scenario: every budget term at the predicted c.....	22
Figure 26 — the Layers page after Reset layers to reconstructor prior, in a fresh GUI with no.....	23
Figure 27 — the same after dragging the 8 km row to 0.40: the other six rows have shrunk prop.....	23
Figure 28 — (a) the Scenario page while layer mode is on: DIMM live, MASS greyed at the deriv.....	24
Figure 29 — the Error terms tab under a layer scenario: the title carries the derived pair (D.....	25
Figure 30 — Dark theme active: the Qt controls restyled dark while the Timeline plot alongside.....	25

1. What this is, and what it is not

This manual is a practical guide to driving the **ao_strehl_timeline** GUI — how to load a night, read the plots, and use the what-if and prediction tools. It deliberately does not explain the underlying physics (the error budget, the Gompertz NGS fit, the PSF model, the synthesized profile). For any of that — what a number means, why a term scales the way it does, how a prediction is built — see the companion document, KAON 1542, which documents the engine the GUI drives. Throughout, a “(KAON 1542 §N)” pointer marks where to look.

The GUI is a front end only. It imports the same engine the command-line tool uses and calls its public functions, so a given set of inputs produces exactly the number the CLI would. Nothing you do in the GUI can change the reported budget physics; the what-if sliders are clearly flagged wherever they are active (§6).

2. Getting started

Launch. Run `keck-ao-estimator-gui`. (The tool ships as the installable `keck_ao_estimator` package: `pip install "[gui]"` provides that command and the CLI’s `keck-ao-estimator`. From a source checkout without installing, `python3 -m keck_ao_estimator.gui.app` still works.) The window opens with an empty plot and the prompt “Set inputs on the left, then press Run.” The left dock holds the input tabs; the right side holds the plots; the bar along the bottom has Run, the Report selector, a status line, and the export buttons. You can also predict before you fetch anything: enable the Prediction tab’s scenario and the field map and Error terms views render with no data loaded at all (§8).

The one required input is a night of data. On the Data tab either point the three file pickers at a local MKWC DIMM / MASS / MASSPRO set, or choose “Fetch by date (MKWC)” and enter a UT date to pull them automatically. Pick the telescope (K1 or K2) and press Run. The status line shows progress; the fetch and the astrometry run off the GUI thread so the window stays responsive.

Run is disabled until the inputs are valid. If a required file is missing or a coordinate cannot be parsed, the Run button greys out and the status line says why (e.g. “Cannot run: DIMM file required” or “NGS offset: star/target coordinates”). Fix the flagged field and Run re-enables itself.

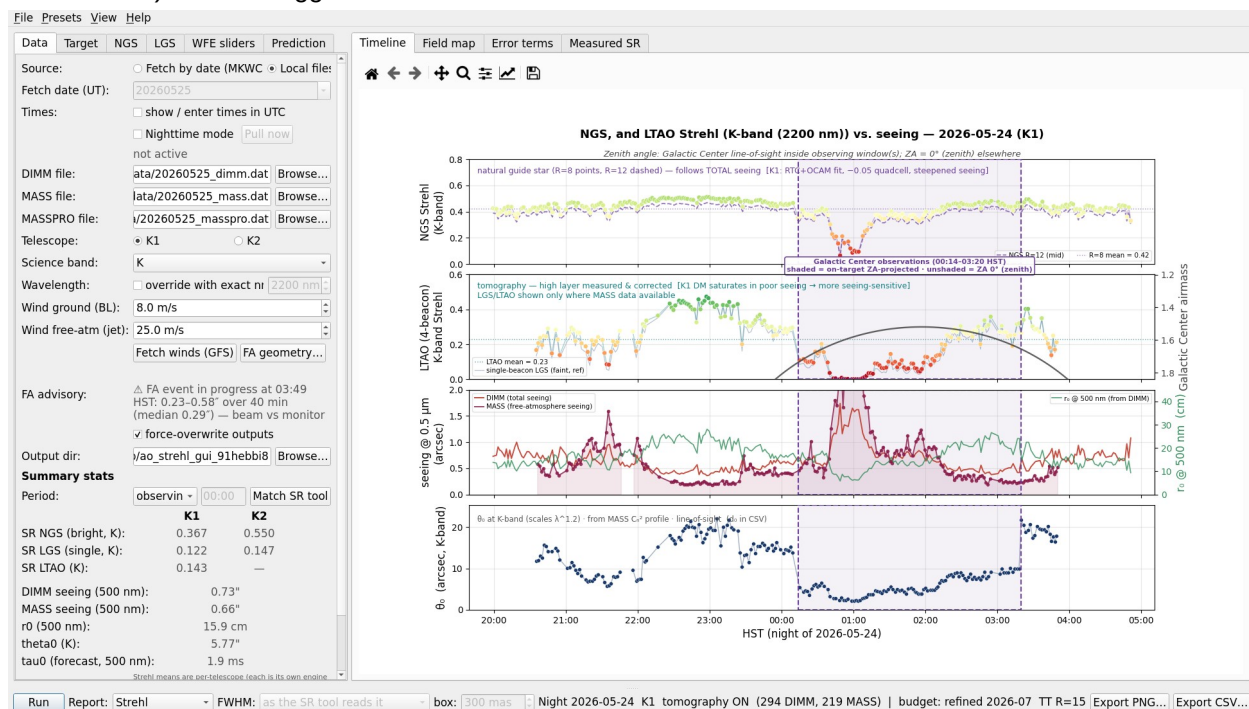


Figure 1 — the main window after a Run: input tabs at left, the Timeline plot at right, and the status bar (night, telescope, budget provenance) along the bottom.

3. The input tabs

The left dock has six tabs. You rarely need all of them: for a standard night, Data + Run is enough. The rest refine the scenario.

Data

Data source (local files or fetch-by-date), the three MKWC file pickers, telescope (K1/K2), science band and an optional exact-wavelength override, the two wind speeds (ground 8 m/s, free-atm 25 m/s by default; they feed the bandwidth term — KAON 1542 §3.4), a force-overwrite toggle for exports, and the output directory.

Nighttime mode. A checkbox for a night still in progress: ticking it auto-fetches tonight's MKWC DIMM/MASS/MASSPRO data and re-runs every 5 minutes, matching MKWC's own publish cadence, so the estimate stays current without you re-clicking Run. It forces fetch-by-date mode on tonight's UT date and greys out the other source controls while on; a status line shows the last and next pull time, and Pull now forces an immediate re-fetch. It only turns on between sunset and sunrise at Keck, and switches itself off within one cycle if left running past sunrise; if the very first pull finds no DIMM file yet (normal early in the evening) it disarms itself rather than repeatedly hammering the archive. Not saved with your config, since turning it on starts background network activity a reopened config should never trigger silently.

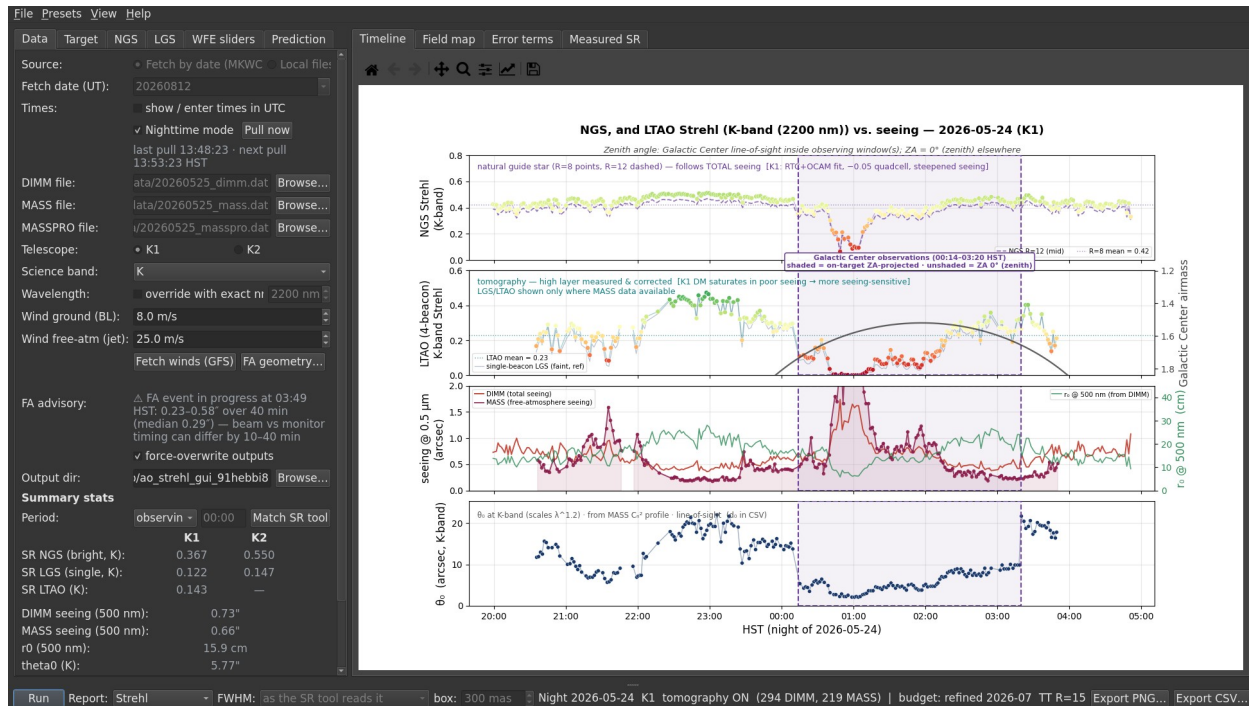


Figure 2 — Nighttime mode enabled: the Data-tab checkbox, the source-selection controls greyed out, the last/next pull status line, and Pull now.

Summary-stats panel. Below the Data-tab controls, a small panel reports the mean NGS, single-beacon-LGS, and (K1 only) LTAO Strehl for both telescopes, plus mean DIMM seeing, mean MASS seeing, mean r_0 and mean θ_0 , over whichever period the field map's Conditions selector is set to (§5) — including, in Nighttime mode, the time of the last auto-pull. Comparing K1 and K2 side by side only makes sense with the STRAP tip-tilt sensor; switch to TRICK (a K1-only sensor with its own science band) and the other telescope's cells read n/a until you switch back.

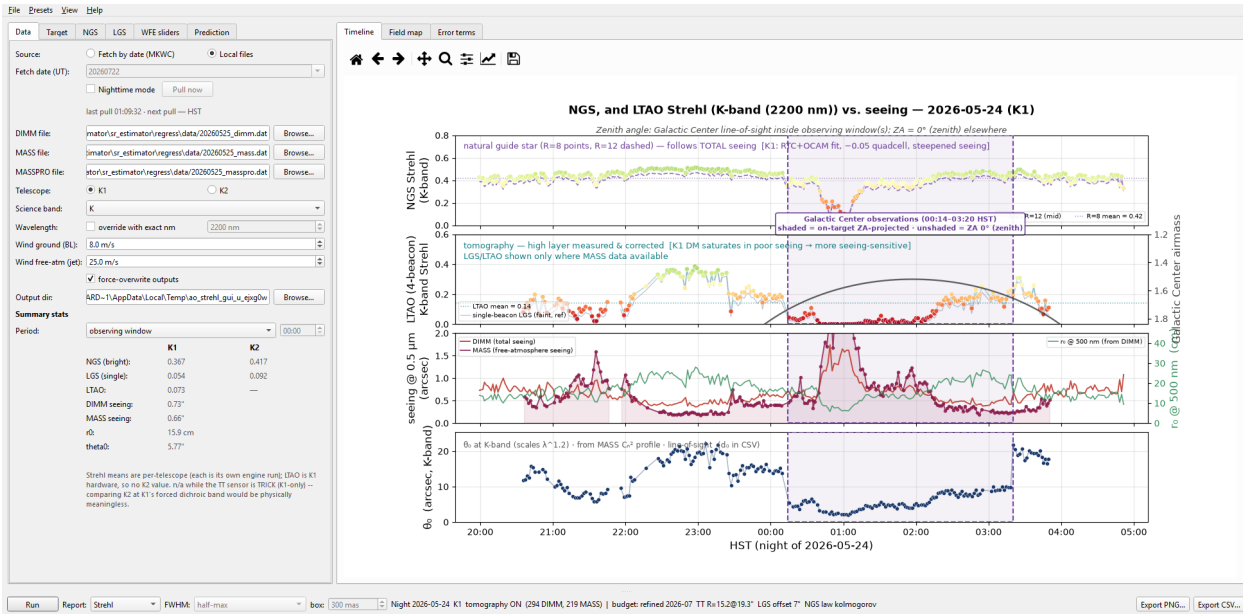


Figure 3 — the summary-stats panel with STRAP selected: NGS, single-beacon-LGS, and LTAO Strehl populated for both telescopes (LTAO is K1-only).

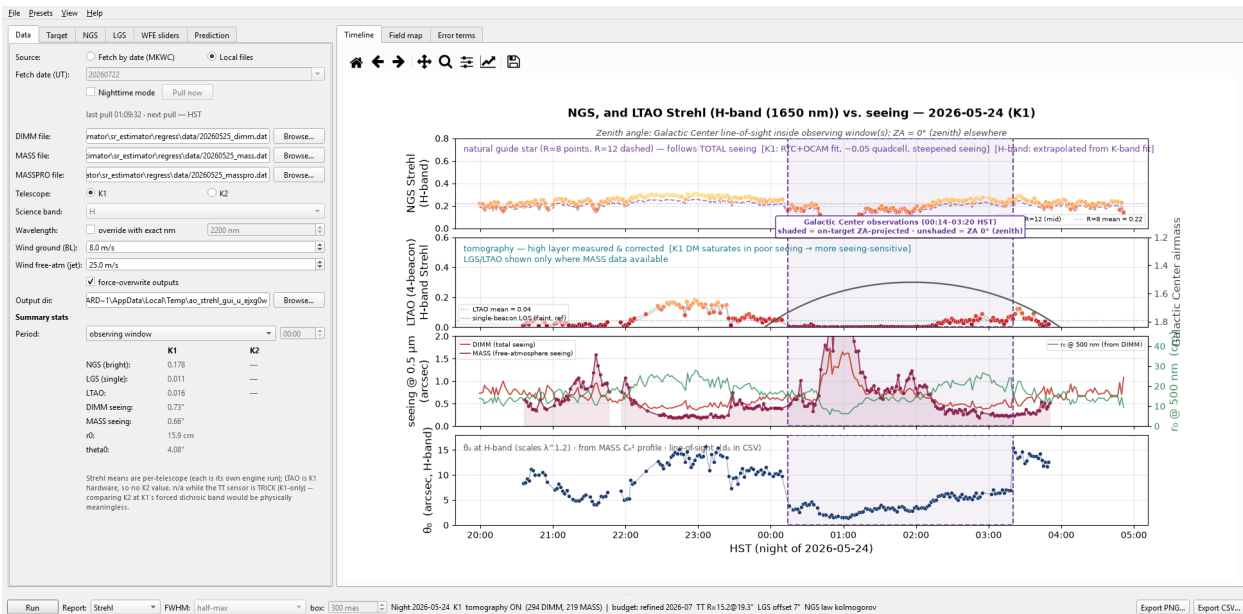
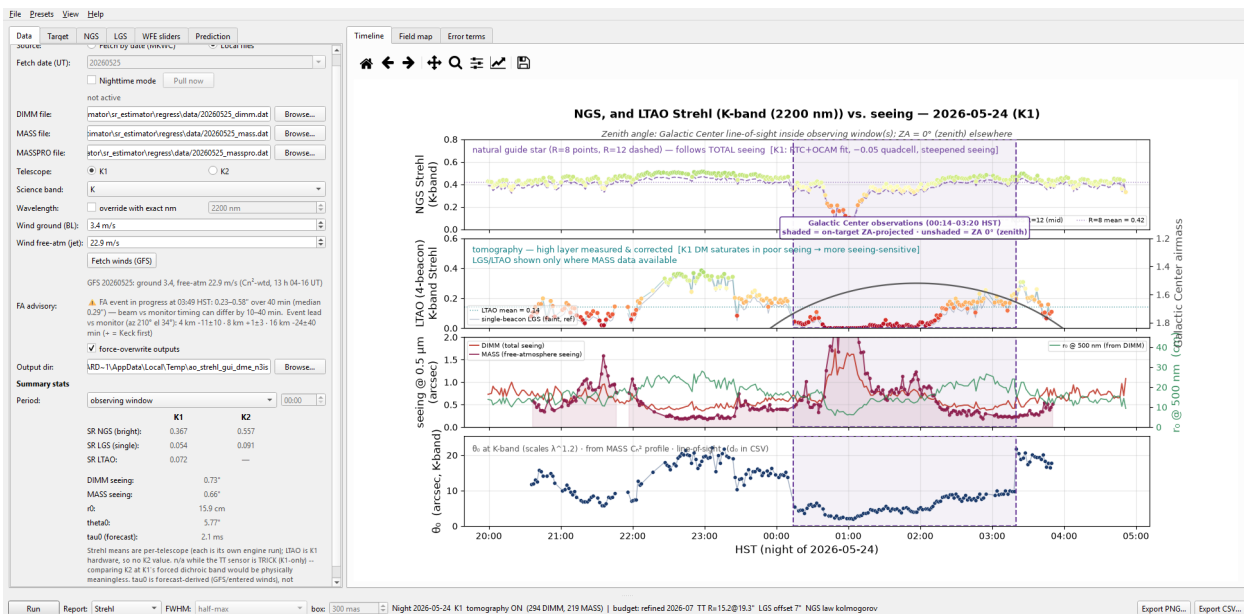


Figure 4 — the same panel with TRICK selected: every K2 cell reads n/a, since a K1-only sensor makes the cross-telescope comparison physically meaningless.

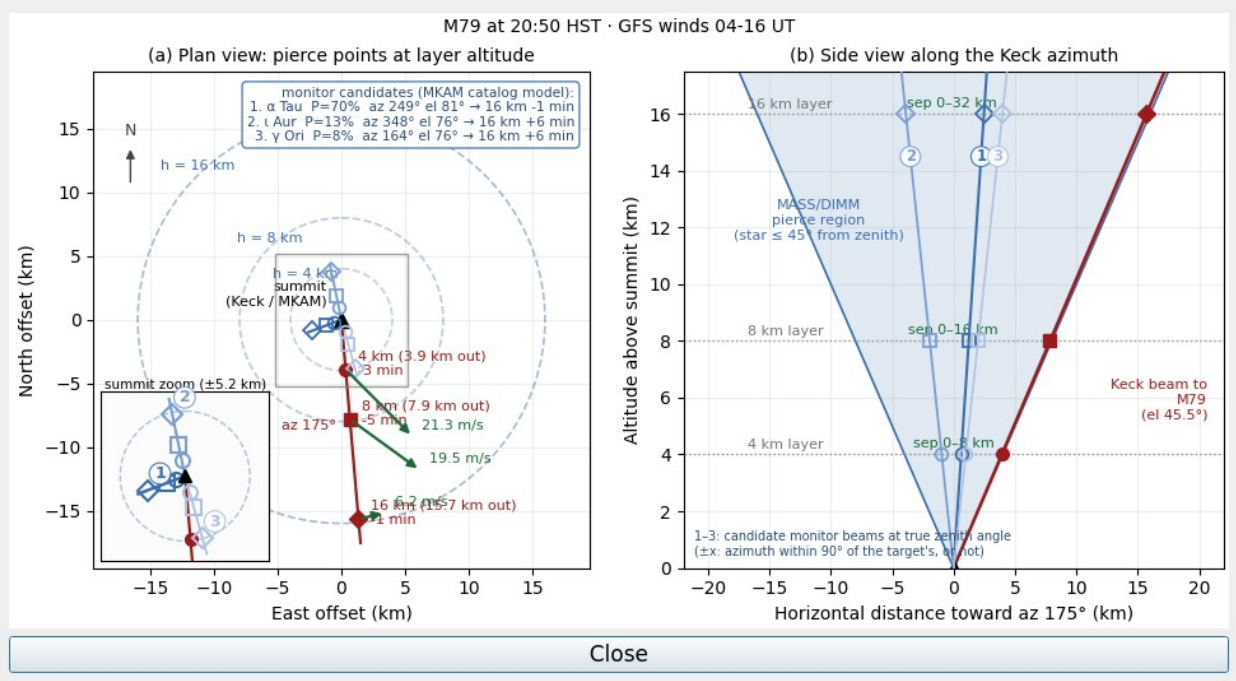
Fetch winds (GFS). One click replaces the 8/25 m/s wind guesses with the night's own GFS analysis, collapsed to the two speeds the estimate actually uses, with the provenance in the status line — the estimate re-runs automatically. A failed fetch warns and changes nothing. A tau0 (forecast) row joins the summary panel below; it is labelled “forecast” because its wind leg is a model product, unlike the measured seeing/ r_0/θ_0 rows (KAON 1542 §9).

FA advisory. A display-only line under the wind controls — it never changes any reported number. It summarizes the trailing 40 minutes of free-atmosphere seeing (median and range) and raises “ FA event in progress” when a genuine spike is underway; with GFS winds fetched and a target set, it adds the expected per-layer lead or lag of your beam versus the monitor — the two pierce a moving layer kilometres apart, so an event can reach one tens of minutes before the other (KAON 1542 §9).



FA geometry... Opens a two-panel picture of why that lead/lag exists for the current target: a plan view of where your beam pierces each MASS layer relative to the monitor’s pierce region, with each layer’s wind arrow, and a side view along the beam. Display-only; it degrades explicitly when there is no run, no visible target, or no winds.

Which star is the monitor watching? The dialog now answers that too. The seeing monitor’s pointing is not logged anywhere, but it works through a fixed catalogue of bright stars, staying near zenith — so the tool ranks the catalogue by how close each star is to zenith at your reference time and draws the three most likely pointings, with their probabilities, az/el, and the lead each implies. A summit-zoom inset makes the near-zenith beams readable, and the lead/lag numbers are pinned to the most probable candidate rather than carrying the full “we don’t know where it’s pointing” spread. Note the reference time follows the Period selector whenever that names a single instant; if your target is below the horizon at that time, the title says so and gives the elevation.



Times and time fields. The Times: checkbox switches every displayed and entered wall time between HST and UTC — plots, window annotations, the Target tab's windows list, and the “specific time” fields — while the engine and saved configs always keep canonical HST. The time fields themselves are now free-typed: “21:35”, “2135”, “935”, or a bare hour, committed on Enter or focus-out; an entry that doesn't parse reverts instead of erroring.

Target

An optional target name and RA/Dec, and a saved list of the night's targets. Save adds or updates the current fields as a named target; Delete removes one; picking from the dropdown reloads its coordinates. Loading a science image on the field map (§5) adds that frame's target automatically. A target is only needed if you want airmass projection onto a specific sky position or a sky-image backdrop; a bare night runs without one.

Coordinate formats. Every RA/Dec field — here and in the “star RA/Dec” offset mode (§4) — takes three formats, detected per field so RA and Dec need not match: self-describing hms/dms (15h49m57.7s / -03d55m16s), colon-separated sexagesimal (15:49:57.7 / -03:55:16, RA in hours, the DS9/pipeline convention), or decimal degrees (237.49 / -3.92).

Target offset. Shifts the typed RA/Dec by an arcsec amount before it reaches the engine — entered the same three ways as any other offset (§4). Use it for an offset guide star or a small dither without hand-computing a new RA/Dec. The typed coordinates stay editable and saveable as a named target; the offset is a layer on top, zero by default, with a live readout of the resolved position. That resolved position is what the airmass calculation, the field map and the offset separations all use — so a target offset moves the target, not the sky: a loaded survey or FITS backdrop stays visually fixed and the target marker moves within it.

Resolving a target. Type a name and click Resolve to query SIMBAD for RA/Dec and, when known, proper motion — your typed name is kept, and a target with no measured proper motion is left alone rather than zeroed.

Proper motion. A row under RA/Dec (also fillable by Resolve) takes PM RA·cosDec and PM Dec in mas/yr, default 0. Unlike the Target offset above, this is applied, not just recorded — the position used everywhere (airmass, field map, every offset) is propagated to tonight's date before anything else. Zero PM changes nothing, so an ordinary target is unaffected; PM is remembered per saved target.

Load starlist.... Reads a standard Keck-format starlist and offers a sortable pop-up of tonight's targets; double-click to pick one. This zeroes any proper motion left over from a previous target, since a starlist carries none, and flags a non-J2000 equinox rather than silently using it.

Picking a target also picks its guide star. For NGS, the guide star is always the target itself — there's no separate choice to make. For LGS, a new target defaults to on-axis too, unless a loaded starlist links a tip-tilt-star candidate to it (the target=<n> field a real starlist can carry, Load starlist above): one linked candidate is adopted directly, the same way as picking a catalogue star on the field map (§5); two or more are ranked by delivered Strehl at the target (the same Rank machinery as the field map, §5) and the best one is used. Resolving a target, saving a new one, or picking a starlist entry with no linked candidate all reset to on-axis — but re-saving a target you've already customized keeps your chosen guide star. Each saved target remembers its own guide star along with its proper motion, so switching between targets never mixes them up.

Its magnitude comes along too. Picking a starlist target also sets its guide-star magnitude — your own value if the starlist has one, a SIMBAD lookup if it doesn't (with a warning if nothing usable turns up) — so a previous target's brightness never lingers on the new one. Starlist target= links now match spaced names too: Keck starlists can only write them with underscores (target=IRAS_23077+6707), and picking now recognizes that as “IRAS 23077+6707”.

A closer look at any starlist entry. Show starlist reopens the picker without re-reading the file. Single-clicking a row (double-click still picks it as the science target) opens a detail panel: RA/Dec and magnitudes, current Az/El, and how far the Moon is with how full it is — all evaluated at an editable “Evaluate at” date and time, which defaults to now and follows whichever of HST or UT the app is currently displaying, labelling itself so you always know which you're typing. A new HA column sits beside RA/Dec and refreshes as you change that time. The Az/El and Moon lines are colour-coded so a problem is visible before you commit to a star: red means below the horizon or blocked by the telescope's pointing limits (the Nasmyth-deck wedge included), amber means vignettted or within 30° of the Moon, red again under 15°, and green means clear.

Editing a starlist without touching the file. Save and Delete on the Target tab now mirror onto a session layer over the loaded starlist: additions go to a .session.lst sidecar in the same Keck-starlist format, and deletions of an entry that came from the original file are recorded by name in a .session_excluded.txt. The picker always shows the merged result. Your actual .lst file is never modified — if you want the changes permanently, fold the sidecar in yourself.

Click a row for details (az/el, Moon separation, evaluated at the time below). Double-click to make it the science target (added to tonight's Targets list). Click a column header to sort.
Evaluate at (HST): 2026-07-28 22:30 Now

#	name	RA	Dec	HA	V	R	K	sep (")	role
1	5	syneng100	08:40:44.20	-19:52:24.1	+9.92h	9.9	—	—	target (lgs)
2	6	syneng101	11:15:10.72	+59:44:54.8	+7.35h	10.1	—	—	target (lgs)
3	7	syneng102	22:04:46.65	-19:55:26.8	-3.48h	10.3	—	—	target (lgs)
4	8	syneng103	14:36:44.41	-11:54:00.3	+3.99h	10.0	—	—	target (lgs)
5	9	syneng104	17:20:22.68	+12:54:44.9	+1.26h	10.3	—	—	target (lgs)
6	10	syneng105	22:08:16.30	+24:53:19.2	-3.54h	10.1	—	—	target (lgs)
7	12	SYN-001	19:22:00.40	-23:26:40.2	-0.76h	—	—	—	target (lgs)
8	13	SYN-002	02:32:57.23	+23:13:56.4	-7.95h	11.6	—	—	target (lgs)
9	14	SYN-003	17:31:57.87	+05:28:28.8	+1.07h	11.8	12.5	9.9	target (lgs)
10	15	SYN-004	04:34:40.93	+59:42:29.5	-9.98h	—	—	—	target (lgs)
11	16	SYN-005	08:10:32.86	-00:09:19.6	+10.43h	11.9	—	—	target (lgs)
12	17	SYN-006	18:51:11.87	+23:21:24.1	-0.25h	14.5	—	—	target (lgs)

syneng100 RA 08:40:44.20 Dec -19:52:24.1 (V=9.9)
Az/El at 2026-07-28 22:30 HST: 264.3° / -60.5° → BELOW HORIZON
Moon separation at 2026-07-28 22:30 HST: 141.2" (illumination 100%, 100% = full moon)

TT star → name rows are tip-tilt-star candidates the list explicitly links (target=) to that science target; they can still be picked directly. Starlists carry no proper motion, so picking a row zeroes the PM fields.

Close

Figure 7 — the starlist picker with a row selected: the detail panel, the HA column, and the “Evaluate at” controls. Red on the Az/El line means this star is inside the telescope’s pointing limit at that time.

Click a row for details (az/el, Moon separation, evaluated at the time below). Double-click to make it the science target (added to tonight's Targets list). Click a column header to sort.

#	name	RA	Dec	HA	V	R	K	sep (")	role
1	1	GuideDemo	15:49:57.7	-03:55:16.0	+2.77h	—	—	—	target (lgs)
2	2	Far_bright	15:50:10.0	-03:55:16.0	+2.77h	—	9.0	—	TT star → GuideDemo
3	3	Close_faint	15:49:58.5	-03:55:20.0	+2.77h	—	16.5	—	TT star → GuideDemo
4	4	Close_bright	15:49:58.7	-03:55:12.0	+2.77h	—	10.0	—	TT star → GuideDemo

Click a row to see its details.

TT star → name rows are tip-tilt-star candidates the list explicitly links (target=) to that science target; they can still be picked directly. Starlists carry no proper motion, so picking a row zeroes the PM fields.

Close

File Presets View Help

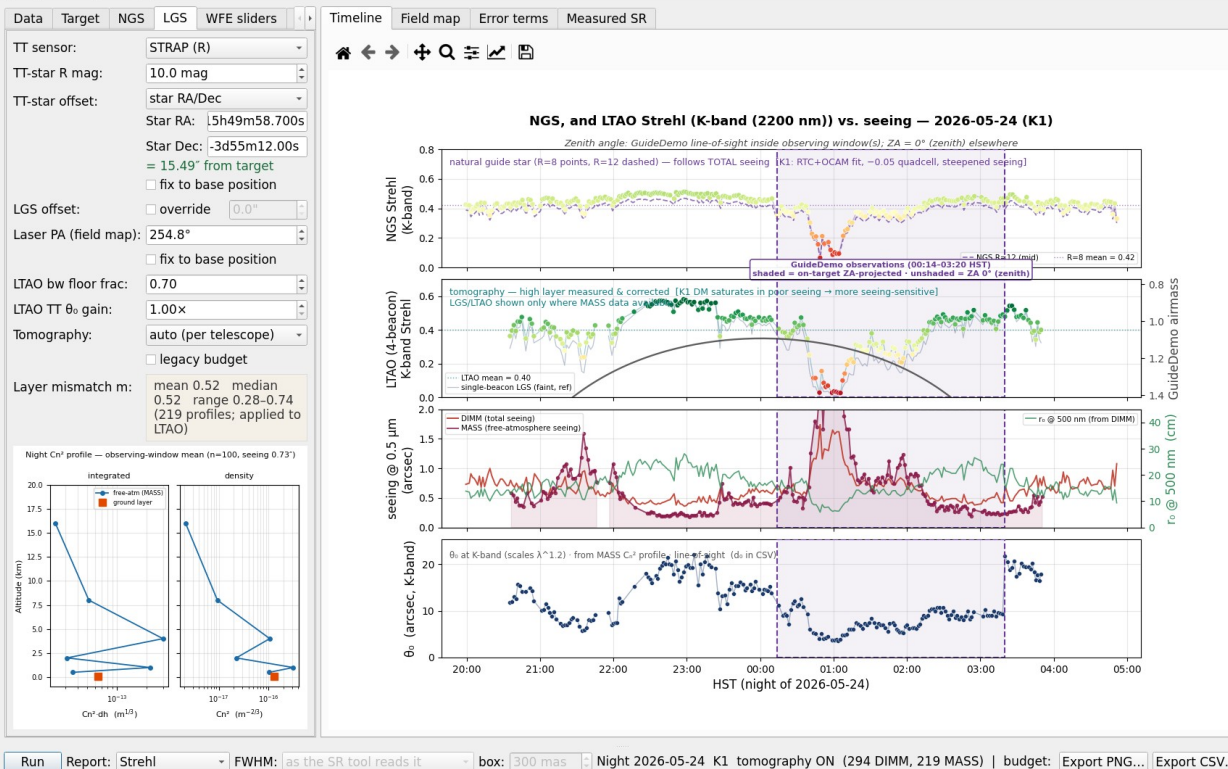


Figure 8 — (a) the Load starlist... picker with three TT-star candidates linked to one target; (b) the LGS tab immediately after picking, with the ranked winner's (Close_bright, R=10.0) offset and magnitude filled in automatically.

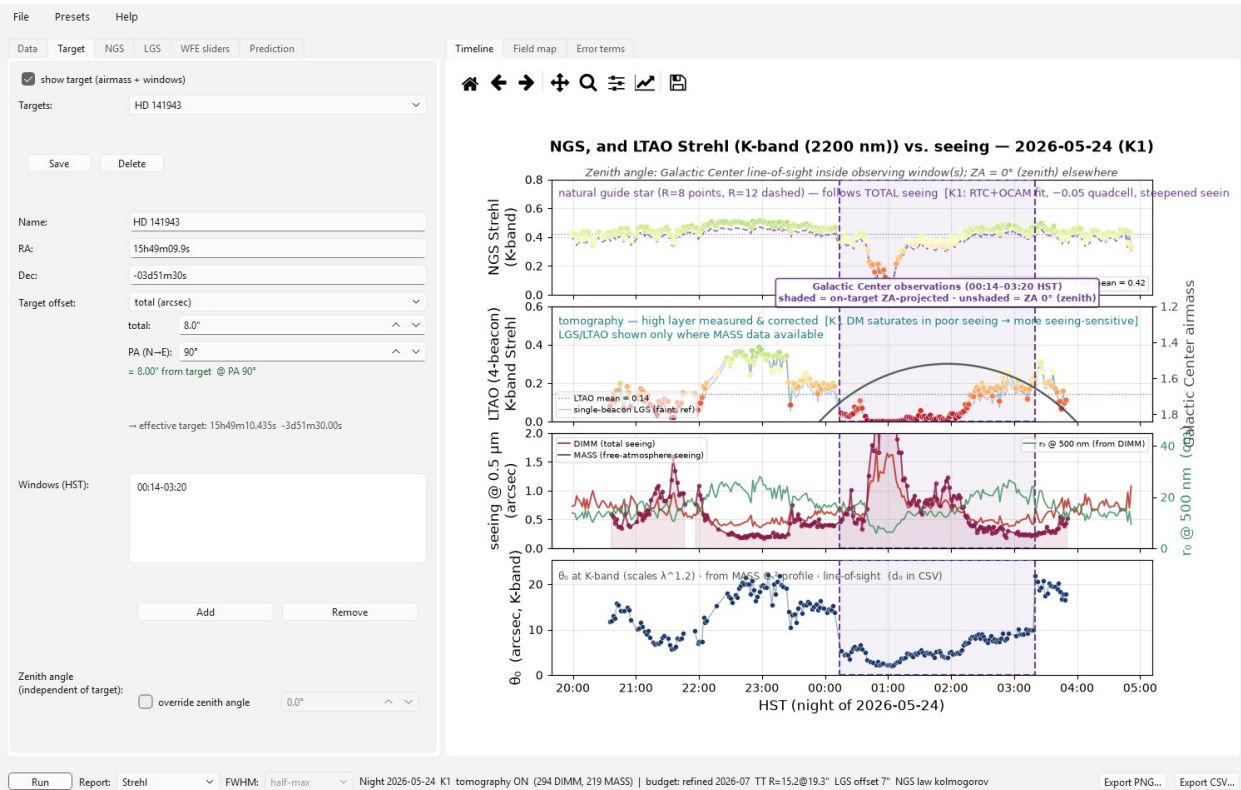


Figure 9 — the Target tab: a saved target list, the Target offset control set to a nonzero shift, and the live resolved-position readout.

NGS

Guide-star magnitudes (bright / faint bounds), the NGS offset (three entry styles — see §4), an assumed θ_0 , the seeing law, and the editable Gompertz fit for the active telescope with a live Strehl-vs-magnitude preview. Editing the fit terms is recalibration, not a what-if — it is flagged as such and does not touch the reported budget (the physics is KAON 1542 §3.2). “Reset fit to telescope default” restores the fitted values.

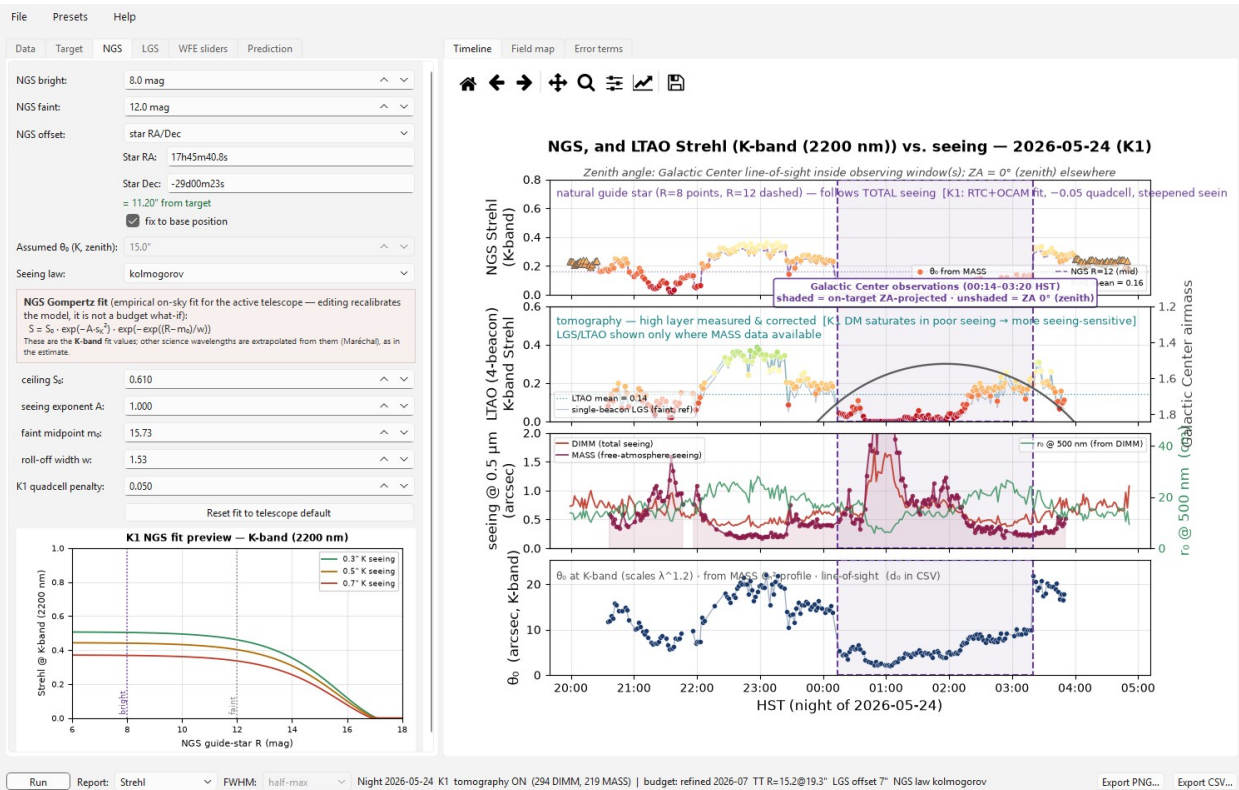


Figure 10 — the NGS tab: guide-star magnitudes, the three-way offset entry, and the editable Gompertz fit with its live preview. Fit edits are recalibration and are flagged, not silent.

LGS

The tip-tilt star magnitude and offset (same three-way entry), the tip-tilt sensor selector (on K1, the R-band STRAP quadcell or the IR TRICK sensor in H or K — TRICK holds tip-tilt to a fainter guide star and swaps the science band via the dichroic; KAON 1542 §3.4 (6)), the LGS laser offset, the Laser PA control (position angle for the field-map laser marker; it now defaults to the campaign-measured 254.8° for K1, replacing the old 225° placeholder — KAON 1542 §3.4 (4)), the LTAO bandwidth floor fraction, the LTAO TT θ_0 gain (see below), a tomography selector (auto / on / off), the legacy-budget toggle, and a live readout of the LTAO layer mismatch. The tab also plots the night's real, as-measured Cn^2 profile for whatever conditions the field map is set to (KAON 1542 §3.5, §3.8).

The LTAO TT θ_0 gain, and why it defaults to 1.1. This spinbox sits directly under the LTAO bandwidth floor fraction and reduces the tilt-anisoplanatism part of the tip-tilt budget — the part that grows with how far your tip-tilt star sits from the science target — in LTAO mode only. It does nothing to a single-beacon LGS estimate, and nothing at all for a star on axis; it matters for far, faint tip-tilt stars, where it can be worth a couple of hundredths of Strehl. The default of 1.1 is deliberately conservative and is not a measured number: the on-sky measurement that would calibrate it has not been made, and how much of it the system actually delivers depends on the real-time controller as well as on the atmosphere. Treat a run at the default as a model prediction rather than a delivered figure, and set the gain to 1.0 if you want the effect off entirely. The physics is KAON 1542 §3.4 (3b). A config saved before this control existed loads with the gain at 1.1.

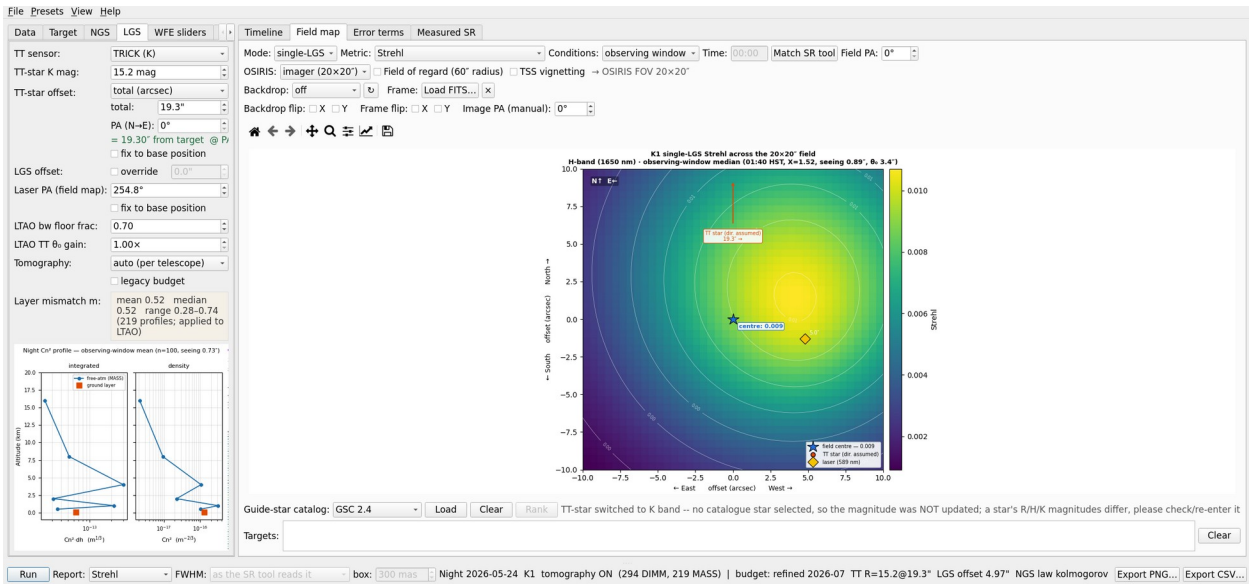


Figure 11 — the LGS tab with *TRICK* (K) selected as the tip-tilt sensor and the night's real Cn^2 profile plotted below.

WFE sliders

One slider per adjustable wavefront-error budget term, in nm at the reference profile, grouped by role (fixed terms, common terms that also project onto the NGS estimate, and LGS-only terms). Rows that do not apply to the current telescope and AO mode grey out automatically. This is the what-if tool; §6 covers how modified budgets are flagged. Each slider has a reset arrow; “Reset all” returns everything to the reference budget.

Budget values: which reference set the sliders start from. A picker above the slider groups chooses between 3_1_3 — the default, and the one you want unless you have a specific reason — and 3_1_1, the values this tool used before July 2026. They differ because the old set had a zenith-angle correction baked in that the pipeline was also applying separately, so everything was penalized twice; KAON 1542 §3.4 has the full story. Predictions changed when this was fixed, and they changed for the better against real measurements, so if you are comparing against numbers produced by older versions of this tool, expect them to disagree. Picking 3_1_1 moves every affected slider through the ordinary what-if machinery, which means the run is tagged MODIFIED BUDGET and every export carries the moved parameters in its provenance — a legacy run can never quietly look like a reference one. Each slider's tooltip tells you where its reference value comes from.

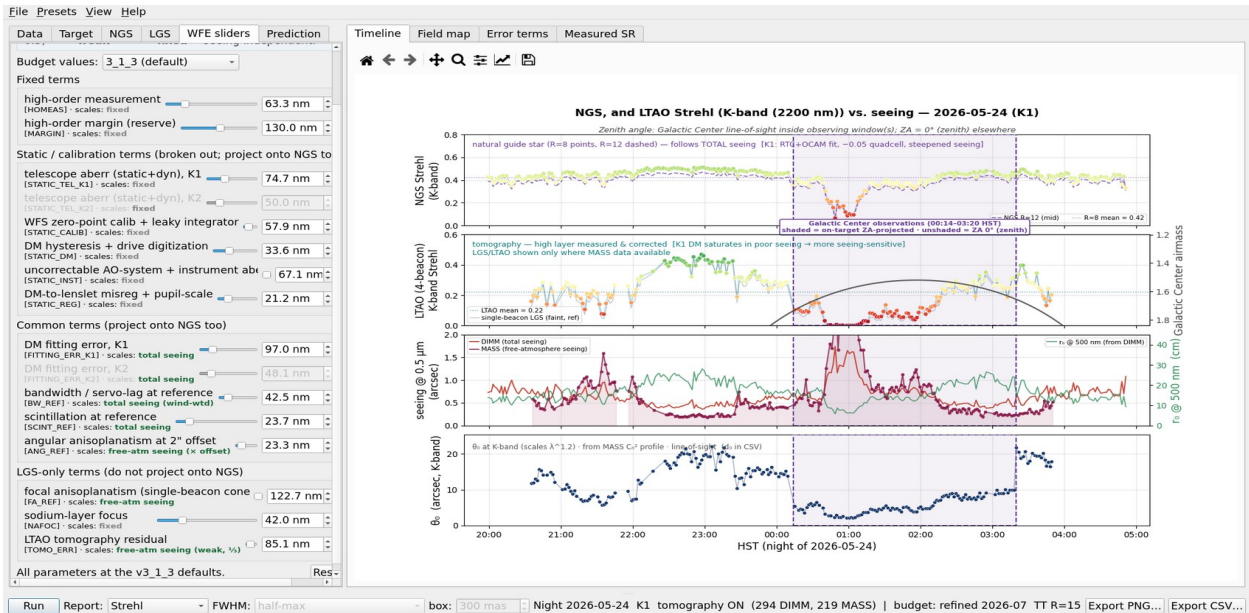


Figure 12 — the WFE sliders tab, with the Budget values picker at 3_1_3. Each slider sets a budget term at the reference profile; the night's seeing scaling is applied on top. Inapplicable rows are greyed for the active telescope and mode.

Prediction

Drives the field map and the Error terms tab from a synthetic atmosphere instead of the night's data (§8, and the physics in KAON 1542 §3.8). Two pages: Scenario — a DIMM/MASS seeing pair with θ_0 and zenith angle, plus three presets that hold total seeing at 0.5" and move turbulence between the ground and the free atmosphere; Layers — the same scenario set as turbulence fractions per layer on the reconstructor's altitude grid. Either way the timeline keeps showing the real night.

4. Entering an offset three ways

The NGS offset (NGS tab) and the TT-star offset (LGS tab) each accept three entry styles from a dropdown, all resolving to the arcsec the engine needs. Use whichever matches the number you actually have:

- **total (arcsec)** — a single separation, plus an optional position angle (North → East) used only to place the marker on the field map. Without a PA the offset is treated as due North and flagged "assumed direction" on the plot.
- **sky-projected ΔRA / ΔDec** — the two on-sky offset components in arcsec; the separation and position angle both follow from them.
- **star RA/Dec** — the offset star's own coordinates; the tool computes the angular separation from the science target (which must also be set). This is for when you have the star's coordinates but not the computed offset.

A green readout under the control shows the resolved arcsec value live. An unparseable coordinate turns the readout red and disables Run until it is fixed.

Fix to base position. Each offset control (TT star, NGS, laser) has this checkbox. By default an offset is relative to wherever the target currently is, so it follows a Target offset (§3) — right for a target-relative dither, wrong for a real guide star or laser point, which does not move on the sky just because you nudged the target. Tick the box and the control's current entry is frozen as an absolute sky position: as the target offset changes, the displayed $\Delta RA/\Delta Dec$ (and, for the laser, the LGS-offset magnitude the engine sees) recompute to keep that real point fixed. No effect unticked, or in star-RA/Dec mode, which is already absolute.

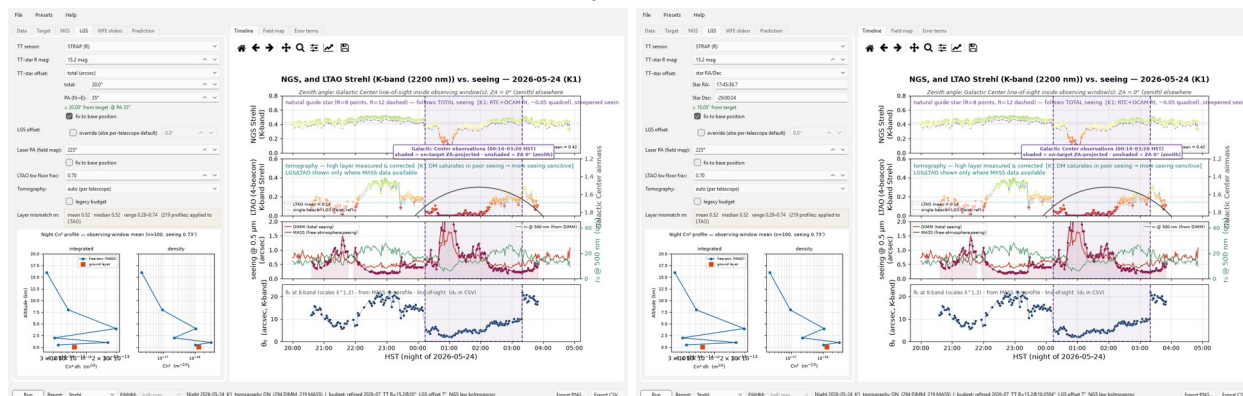


Figure 13 — the TT-star offset (LGS tab) with “fix to base position” ticked, entered as a total + PA (left) and as star RA/Dec typed in colon-sexagesimal (right); the readout shows the resolved separation.

5. Reading the plots

The right side has three plot tabs.

Timeline

The main output: predicted Strehl (or FWHM — see the Report selector, §6) across the night for NGS, single-beacon LGS and LTAO, with the seeing, r_0 and θ_0 panels below. Line breaks are real gaps in MASS coverage, not interpolated. This tab always shows the real night, even while the Prediction tab is driving the field map.

Field map

Performance across the science field at one instant, rather than on-axis across the night. Choose the AO mode (NGS / single-LGS / LTAO), the metric (Strehl or FWHM in either convention), and the atmospheric snapshot (observing-window median, whole-night median, or a specific time). The science target, offset/TT star and laser are marked; the target carries its science-direction value, which reproduces the timeline number for the same instant. Two overlays make it an on-sky planning tool: a field-of-regard toggle expands the view to the 60" LGS patrol circle,

and a sky-image control draws a DSS/2MASS backdrop — or a local FITS science frame, which defines the field from its own header pointing and lets you read performance at the positions of real candidate guide stars. The physics of the off-axis re-evaluation is KAON 1542 \$9 (field map) and \$3.8.

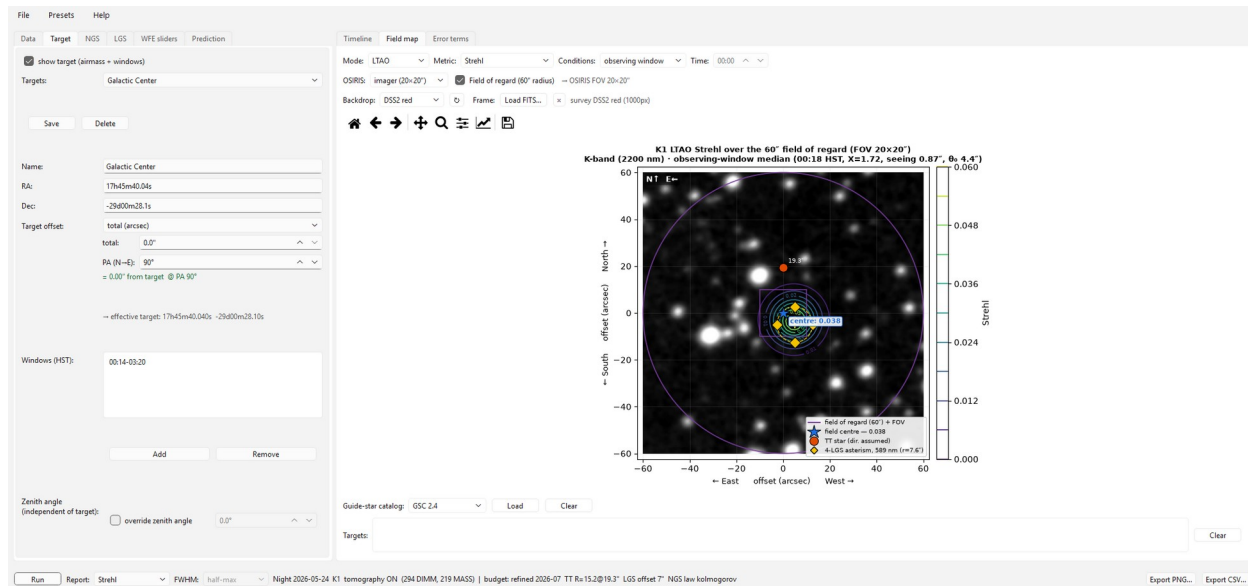


Figure 14 — the field-of-regard view with a DSS2 sky backdrop: the 60" patrol circle, the science-FOV box, and contour-only performance drawn over the image so candidate guide stars stay visible; the Guide-star catalog combo sits in the control row.

Several targets on one map. The grid origin is the field centre — where the laser is pointed and where the on-axis value is reported. You can drop any number of science-target markers at offsets from it; each is annotated with its own predicted Strehl or FWHM at that position and listed — with its own RA/Dec and its offset from the field centre — in the Targets panel below the map. Right-click the map for a menu to drop a target at the clicked point, or to put the laser, TT star or NGS star there. The usual case is two or more targets a few arcsec apart in one science field with the laser on the field centre — read the delivered performance on each at once. Each value is evaluated with the same model that builds the map, so a target's number always agrees with the heat map under it. Dropped targets are saved with the config.

Moving the target itself: “Redefine field centre here.” The same right-click menu has an action that is easy to confuse with “drop a science target here” and does something quite different. Dropping a target leaves a lightweight marker at an offset from the field centre. Redefining the field centre moves the currently-selected target's own RA/Dec to the point you clicked — a real re-pointing, which is what you want when you are nudging where the telescope will actually look. Clicking repeatedly updates that same target rather than piling up new ones. As with the Target-tab offset, a loaded sky image stays put and the target marker moves within it.

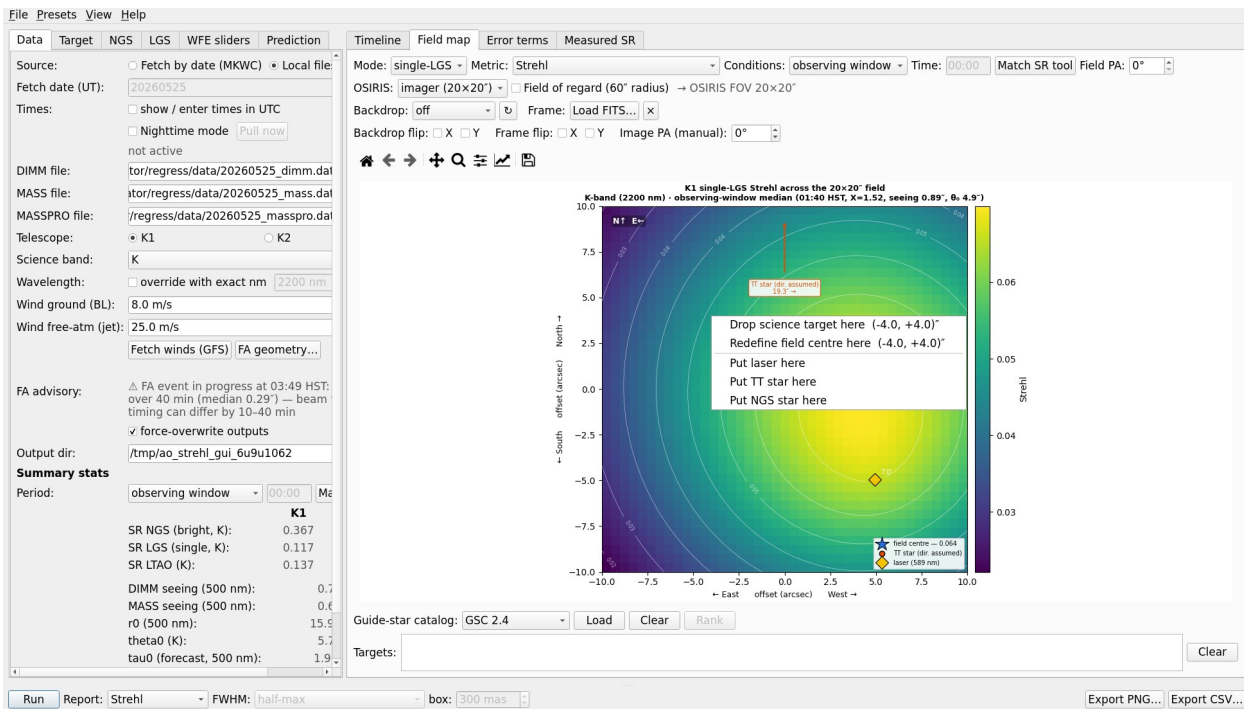


Figure 15 — the right-click menu with both actions visible — dropping a target and redefining the field centre, at the same clicked offset. (Composite of two widget renders: a Qt menu is its own window and cannot be captured in the same grab.)

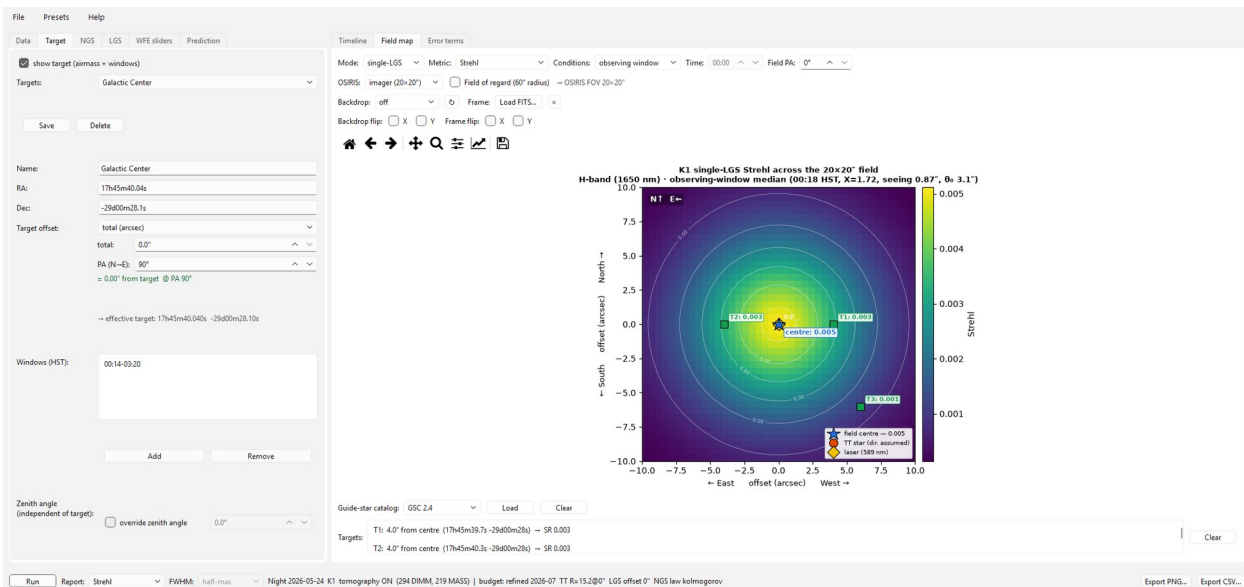


Figure 16 — the field map (K1 single-LGS, science field) with three science-target markers dropped at offsets from the field centre, each now listing its own RA/Dec alongside its predicted Strehl and offset in the Targets panel below the map.

Picking guide stars off the map. The Guide-star catalog combo (GSC 2.4, 2MASS, UCAC4, PanSTARRS DR2, Gaia DR2) plus Load queries that catalogue around the field centre out to the 60" field of regard and plots the candidates, drawn as unfilled rings — never solid, so a star loaded under a sky backdrop is never hidden under its own marker — sized by brightness in the tip-tilt sensing band, bigger meaning a better guide star. Stars past the sensor's practical limit (about R 17.5 for STRAP, 15.5 for TRICK H/K) or with no usable magnitude are small and hollow, and switching the TT sensor re-sizes everything to the new band. Left-click a star to inspect it: it is ringed and its magnitudes, including the sensing band, print in the status line without changing any control. Right-click for "Set as TT star" or "Set as NGS star", which records the star's real RA/Dec (not a delta from the target, so it stays correct if you move the target afterward) and its magnitude in the sensor's band — so you can take a real candidate and read its predicted performance immediately. Switch the TT sensor between STRAP and TRICK afterward and the tracked star's magnitude re-derives automatically in the new band (R, H and K are never the same number); editing

the magnitude field by hand stops that tracking, and the status line tells you if no catalogue star or no usable magnitude is available. Right-clicking bare sky with no star under the cursor (“Put TT/NGS star here”) still uses a simple offset, since there is no real star to record. Magnitudes are the catalogue’s own where it has the band, otherwise converted from a published colour transform and flagged as estimated; near-IR H/K is only ever taken from real near-IR photometry, never synthesised from optical colours (KAON 1542 §9). Along with the sky backdrop, this is the one feature that needs a network connection. (Local FITS is not the only option here — a PNG with an embedded sky position loads the same way, useful for a light example image instead of a large FITS.)

Ranking candidates instead of checking one at a time. Once a catalogue is loaded, Rank scores every candidate as the guide star and orders them by the delivered performance they’d actually give at your target — not just how bright or close they are. The top 3 are badged on the map, and the full sortable table (with an exclusion reason for stars that don’t qualify) opens separately; double-click a row to adopt that star, or single-click to highlight it on the map (and vice versa — clicking a star on the map highlights its table row). Re-click Rank after changing catalogue, TT sensor, or Mode/Metric; dragging the laser or an offset does not clear it, so you can keep comparing candidates while you work.

Watch for the reddening warning. In a dusty field, a star known only in the infrared can look bright in a fabricated optical magnitude that is wrong by many magnitudes. The tool catches this and marks the star amber (“⚠️ reddened ... verify”) rather than letting it rank first on a number it can’t trust — switch to the TRICK infrared sensor and the same star is fully usable, since infrared photometry isn’t fooled by dust the way an estimated optical magnitude is.

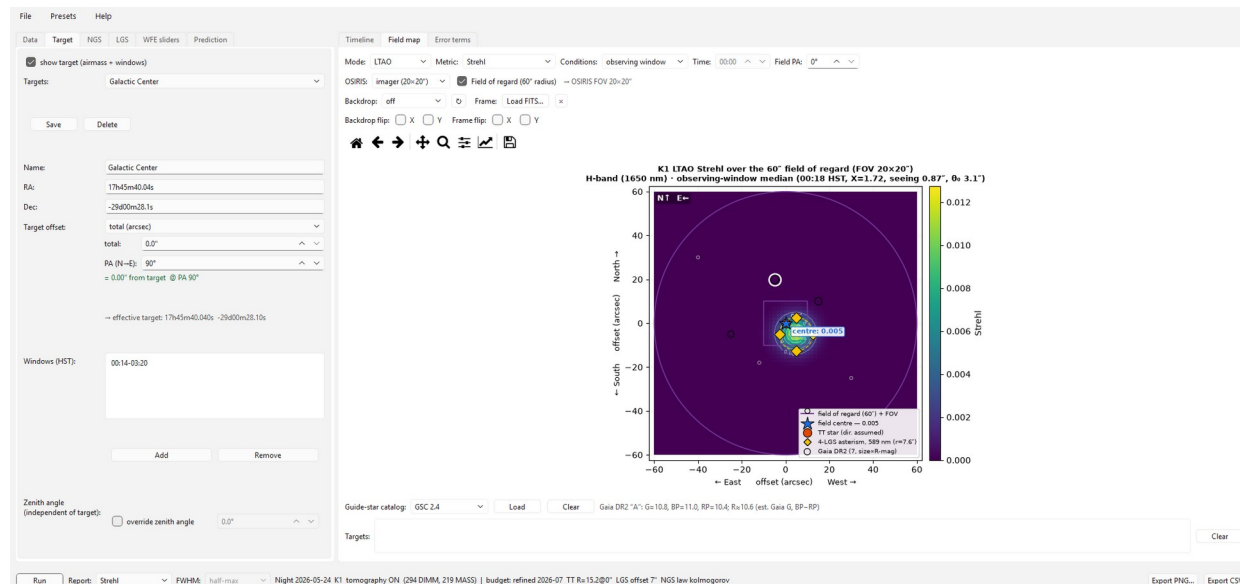


Figure 17 — the catalogue overlay with the current unfilled-ring, size-graded markers (GSC 2.4, field of regard); the status line shows a TT-sensor band switch with no catalogue star currently tracked.

Rotating the view: Field PA. A Field PA control spins the whole displayed field — backdrop, markers, the field-of-regard and FOV outlines, and the N/E compass, all together — to a chosen position angle. It’s a display setting only: nothing about the underlying performance numbers changes, it just lets you line the view up with a rotated real instrument frame or FOV.

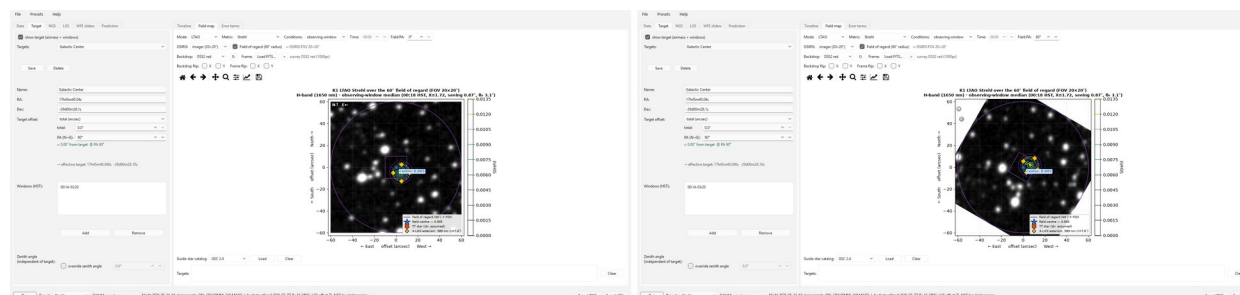


Figure 18 — the same view at Field PA = 0° (left) and 60° (right): the backdrop, outlines and compass rotate together; the numbers underneath do not change.

Fixing a mirrored frame: Backdrop/Frame flip. If a loaded survey image or local FITS frame looks mirrored, tick “Backdrop flip” or “Frame flip” (X and/or Y, independently) to correct it on screen. This only changes how the image

is drawn, not the file or its astrometry. If instead the image is rotated wrong relative to real stars, Image PA (labelled “MANUAL OVERRIDE”) rotates just the backdrop/frame, leaving the catalogue and markers where they are, so you can turn it by eye until it lines up; whenever it is non-zero a red “⚠ IMAGE MANUALLY ROTATED” banner appears on the map as a reminder. This is different from Field PA (§5), which rotates everything together purely for display and never needs a warning.

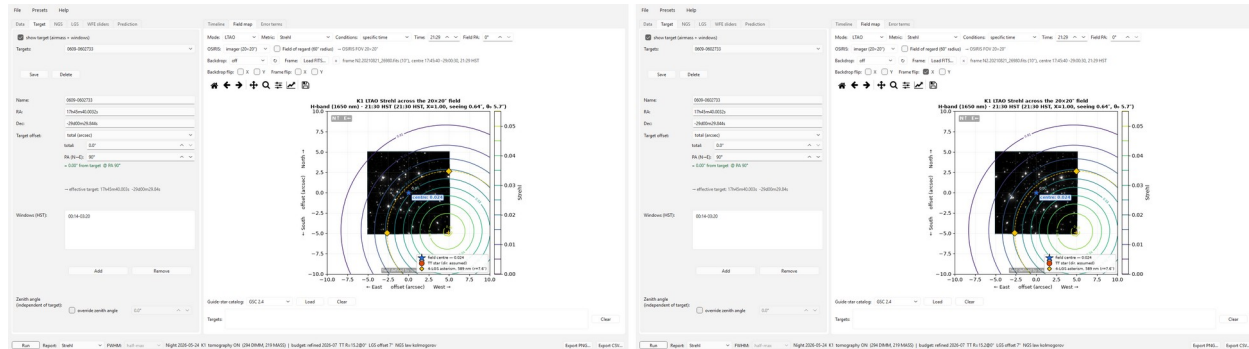


Figure 19 — a loaded frame unflipped (left) and with Frame flip: X ticked (right).

Also new: mouse-wheel zoom on the field map, centred wherever the cursor is.

Error terms

The per-sample error-budget breakdown for the laser modes — each term’s nm contribution across the night, and the stacked percent-of-variance view. Useful for seeing which term dominates when a prediction looks surprising (the terms themselves are KAON 1542 §3.3). Under an active Prediction scenario (§8) this tab switches to a snapshot view instead: grouped bars of every budget term at the scenario’s conditions, single-beacon LGS beside LTAO, with the totals and the resulting Strehl per mode in the title; toggle the scenario off to get the night’s time series back.

6. Measuring real frames: the Measured SR tab

A fourth plot-side tab, Measured SR, measures delivered Strehl and FWHM straight from NIRC2 and OSIRIS science frames — the summit IDL Strehl tools reborn inside this GUI, validated against their own golden outputs. The physics, the port’s provenance, and its validation live in KAON 1542 §10; this section is the driver’s guide.

Loading and measuring a frame. Point Frames at a directory and either run a numbered sequence with Measure (first image / count / backgrounds, the IDL GO! flow) or double-click any FITS in the list. Frames route by their own headers — a NIRC2 frame uses the bundled K2 superflat/supermask calibration automatically, an OSIRIS frame uses that fork’s reduction — and a foreign instrument is refused by name. Find the star with Autofind (brightest pixel) or click it yourself: the PICK ZOOM magnifier follows the cursor and locks on click (Figure 21). The MODEL PSF and MEASURED STAR cutouts sit side by side, and every measurement appends an aligned measured/predicted pair of log lines.

Photometry controls. The photometry, sky-annulus, and peak-finding radii are the IDL tool’s own, with its ordering guard. Two crowded-field aids are worth knowing: Robust sky replaces the annulus mean with a σ -clipped median when neighbours contaminate it, and Pick sky uses an empty patch you click instead. Auto aperture chooses the photometry radius from the curve of growth — it stops where the star’s flux is all collected, or backs off where a neighbour’s wings start entering — and keeps the Strehl normalization self-consistent at whatever radius it picks. An amber CROWDED tag, a red UNPHYSICAL SR tag, and per-measurement $\pm$ S error bars keep the output honest.

Reading the results. The Results box shows the frame’s OBJECT and RA/Dec (Set as target pushes them to the Target tab), the measured SR/FWHM/WFE, and — when an estimate is loaded — the predicted value at the frame’s own UTC and wavelength, with the Δ . Match SR tool drives the summary-stats and field-map selectors to the frame’s time and wavelength so every panel talks about the same instant.

Measuring a whole field. The Field map sub-tab measures many stars at once: Measure field auto-finds the brightest candidates (set Stars, or leave it at Auto to keep accepting stars while their error bars stay tight), measures them asynchronously with a progress counter, and drops rejects — saturated, unphysical, or inconsistent with the rest of the field — with logged reasons, backfilling deeper candidates until enough are accepted. Add star by click for anything it missed. The map colours stars by SR or FWHM, rings the best star in gold, draws the downhill performance gradient as a gold arrow, and — with four or more stars — reports an isoplanatic angle fitted from your

own data (Figure 22), with a 500 nm companion value that sits directly next to the MASS/DIMM θ_0 in the summary stats. Pop out opens the map in its own resizable window. The tab's inputs persist in saved configs.

Checking and pruning a measured field. Click any star on the map — embedded or popped out — to select it: the Results block and cutout switch to that star, and a Reject star button arms. Rejecting it removes it and recomputes every field statistic (peak, mean, gradient, θ_0), with a line in the log saying you did it. The popped-out map also has a “Show image (hollow markers)” checkbox that draws the real reduced frame behind the markers and switches them to hollow rings, so you can check a measurement by eye against the star underneath it. You can add a star by clicking directly in the pop-out too, without switching back to the Image sub-tab.

Stars near the edge of the detector. Watch for an EDGE tag. A star close enough to the edge that its sky annulus runs off the array has a truncated curve of growth, which fools the auto aperture into settling too small and inflates the Strehl — a real edge star read 0.50 that way against a physical 0.24. When more than 5 % of the annulus falls off-array the tool skips the optimizer, uses the fixed radius, and tags the result. Interior stars are unaffected.

Crowded fields: the EE aperture correction. If you have ever wondered why this tool's Strehl disagreed with the summit tool on the same star, the answer is aperture convention: a small photometry radius misses halo flux that the diffraction-limited reference does not lose at the same rate, which reads high by 1.6–3.3×. Tick EE aperture correction (field only) — it needs Auto aperture, and roughly doubles the time a field measure takes. It measures a full-radius companion for each star, calibrates the field's own correction factor h from the uncrowded ones, and applies it. h is calibrated per field and never carried between fields. It is also never silent about what it did: the fitted h appears as a badge beside the field stats, and the log tells you whether the calibration was clean, loose (on a field with no genuinely uncrowded companion — exactly the case this exists for), or absent, in which case your values stay in the small-aperture convention and it says so.

Crowded fields, part two: PSF-fit neighbour subtraction. The EE correction above, Robust sky and Pick sky all fix the sky. None of them touches a star sitting inside your photometry aperture, which in a dense field is the bigger error. Tick PSF-fit neighbour subtraction (off by default) and the tool builds a PSF model out of the frame's own isolated stars, fits it to your target and to every catalogued neighbour close enough to matter, and subtracts the neighbours — never the target — before measuring exactly as it otherwise would. Same aperture, same sky, same arithmetic; only the pixels the neighbours occupied are different. With the box unticked nothing about your numbers changes at all.

Read the tag, and know which way it is wrong. The result carries a direction, not just an error bar. Below Strehl 0.30 — the validated range — a cleaned number errs low, so you can treat it as a floor on the real performance. Above 0.30 the sign flips: you will see a PSF-CLEAN ABOVE VALIDATED SR tag, and that number should be read as an upper bound, not a measurement. On a moderately crowded field expect the cleaned value to lean high by roughly +0.10 in Strehl. Note also that on any star where cleaning succeeds, Robust sky is ignored in favour of the plain annulus mean — the two corrections remove the same light twice, and using both roughly doubles the leftover bias. The log records that substitution per star, and CROWDED still shows either way, because it describes the field, not whether the measurement was corrected.

When it declines — which, so far, is on every real NIRC2 frame. The model needs at least four isolated, well-exposed stars in the frame. Sparse fields do not have four stars; dense fields do not have four isolated ones. Across a sparse standard field, M92 and the Galactic Centre, none has supplied them, so the feature declines, says why in the log, and leaves your default measurement untouched. You may also see it decline after building a model, if that model predicts more neighbour light than the aperture can physically hold — that check runs once per frame, so it costs one log line rather than a complaint on every star. Either way nothing is silently changed.

Finding the tip-tilt star, and reading archive filenames. The TT star button works out how far the tip-tilt star sat from the field centre from the frame's own AO headers — the delivered separation, not the commanded one — then lists catalogue stars at that radius, within $\pm 5''$, with the best magnitude available. It deliberately does not tell you the direction: that is not recoverable from a single frame. Separately, if you are working from an archive download whose filenames are all date-and-seconds, tick Show native filenames above the frame list and the tool relabels the list from each file's own header, so it matches your night log again. Nothing on disk is renamed.

The Image log. Alongside the free-text log, every measurement now also fills a structured row: time, frame number, target, guide-star magnitude and where that magnitude came from, RA/Dec, measured and predicted Strehl and FWHM with their differences, filter, airmass, pixel position, Az/EI, and LBWFS and DIMM/MASS seeing. Pop out the log and you get an Image log tab — a sortable table with Delete selected and an Export CSV... button. Export only happens when you ask. The Append to log checkbox beside Measure turns the whole thing off for throwaway re-measurements. Measure the same frame twice and the run pauses to ask whether to append, overwrite, or skip, with a box to apply your answer to the rest of the run.

Time (UTC)	Frame #	Target	Guide mag	Guide mag source	RA	Dec	Measured SR	Predicted SR	Δ SR	Measured FWHM (mas)	Predicted
2026-01-31 06:50:38	i260131_a028002	M79	10	target list entry	81.044125	-24.52425001	0.395	0.569	-0.174	48.982	---
2026-01-31 06:50:38	i260131_a028002	M79	10	target list entry	81.044125	-24.52425001	0.406	0.569	-0.163	51.939	---
2026-01-31 06:50:38	i260131_a028002	M79	10	target list entry	81.044125	-24.52425001	0.403	0.569	-0.166	48.612	---
2026-01-31 06:50:38	i260131_a028002	M79	10	target list entry	81.044125	-24.52425001	0.439	0.569	-0.129	49.721	---

Figure 20 — the popped-out log on its Image log tab, with the Guide mag source column showing where each magnitude came from.

Where the guide-star magnitude comes from. This one is worth knowing, because it used to be wrong in a quiet way. The log used to record whatever was in the TT-magnitude spinbox, which on a frame you loaded from disk is usually just the default — so the log was mislabelled, and the prediction it was being compared against had been computed for the wrong star. The magnitude is now resolved from real evidence in order: this frame's TT-star catalogue match, then a starlist-linked TT star (laser frames only), then the target-list entry's own magnitude, then on NGS frames the target itself, and only as a last resort the spinbox. The source is logged beside the number, and if it disagrees with what your loaded prediction assumed by more than 0.3 mag, the log says so. It warns rather than silently re-running underneath you.

If the FWHM comes back blank. A dash rather than a number in the FWHM column means the radial-profile fit found no half-max crossing — usually a very bright or saturated star, which is exactly what autofind tends to land on in a cluster field. That is reported as absent rather than as a number, and the Δ FWHM is dropped with it, so nothing downstream inherits a bad value. Click a cleaner star instead. One caveat: the on-screen MEASURED FWHM box and the free-text log still show the raw sentinel (a negative number) for a failed fit — only the Image log and the exported CSV blank it.

Driving the tool from a script. The duplicate-frame prompt is a modal dialog. That is what you want interactively, but an automated run with nobody to answer it will hang, so set the duplicate-handling policy in advance if you script the tool.

The screenshot displays the 'Measured SR' tab in the software. The left sidebar contains various configuration options. The main panel shows the 'Frames' tab with a list of frames and a 'Pick sky' button. The 'Photometry' section shows various radii and a 'Robust sky (o-clip)' checkbox. The 'Results' section shows the measured and predicted SR and FWHM values. The 'Log' section shows a detailed log of the measurement process.

Figure 21 — the Measured SR tab mid-pick on an M79 OSIRIS frame: the PICK ZOOM magnifier locked on the clicked star, the MODEL PSF cutout, the PSF-fit neighbour subtraction checkbox in the Photometry group, and the measured/predicted log below.

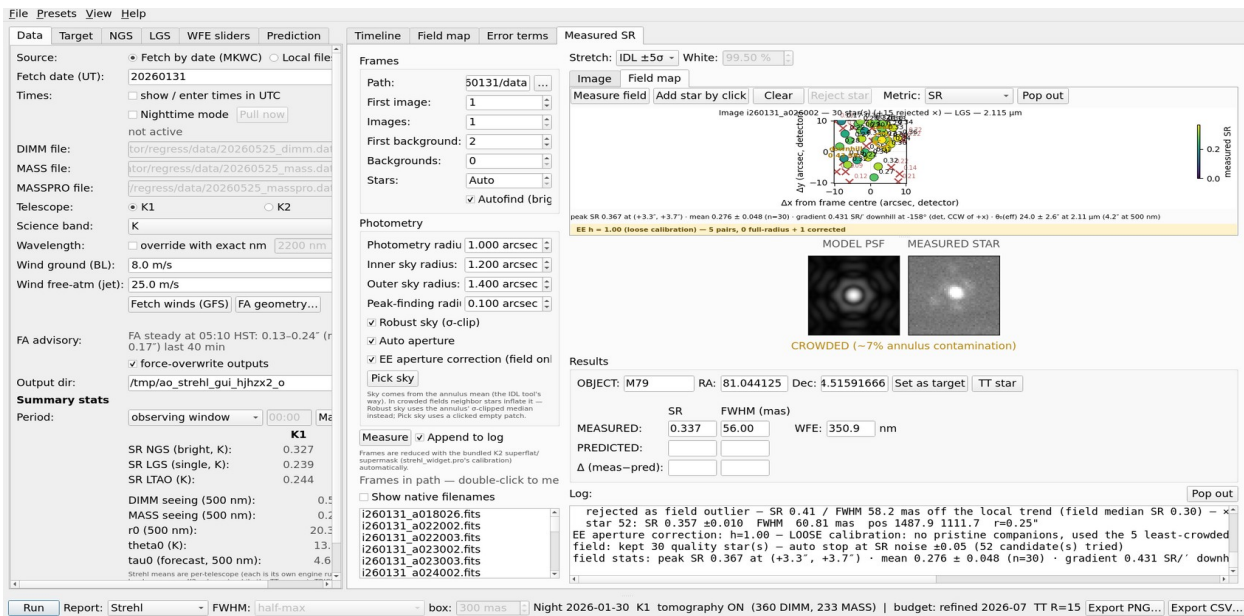


Figure 22 — the measured field map on the same frame with EE correction on: 30 quality stars kept and 15 rejected, the Reject star control, the gold peak-star ring and downhill gradient arrow, the field-stats line with the data-fitted $\theta_0(\text{eff}) = 24.0 \pm 2.6$, and the EE h badge — which reads “loose calibration” here, because this field had no strictly-clean companion to calibrate from.

7. What-if, reports, and honest provenance

Live updates. After the first Run, most edits update the plot in about a third of a second without a full re-fetch: NGS and TT-star settings, the LGS offset, the WFE sliders and the winds recompute on the cached night. Changes that alter the night geometry itself — telescope, wavelength, zenith, tomography — re-prepare in the background and take a moment longer.

The Report selector (bottom bar) switches the Timeline between Strehl, FWHM, and both at once. In an FWHM view the “FWHM” combo beside it picks which convention(s) to draw. There are now four, and the first — “as the SR tool reads it” — is the default: it is what the Measured-SR tab (§6) would actually read off this PSF, so it is the one to quote when you are comparing a prediction against a measurement. The other three (half-max, and the two Gaussian-fit variants with or without a free background) are still there, and the CSV always carries all four. KAON 1542 §3.7 explains the differences.

Modified budgets are never silent. Whenever any WFE slider is off its default, a red MODIFIED BUDGET indicator appears on the plot, and every export records the exact overrides — `budget_overrides=NAME=value` in the CSV provenance and a footer on exported PNGs. A what-if figure therefore can never be mistaken for a reference-budget figure once it leaves the GUI. The Presets menu’s “Reference budget” entry (Ctrl+0) clears every override at once.

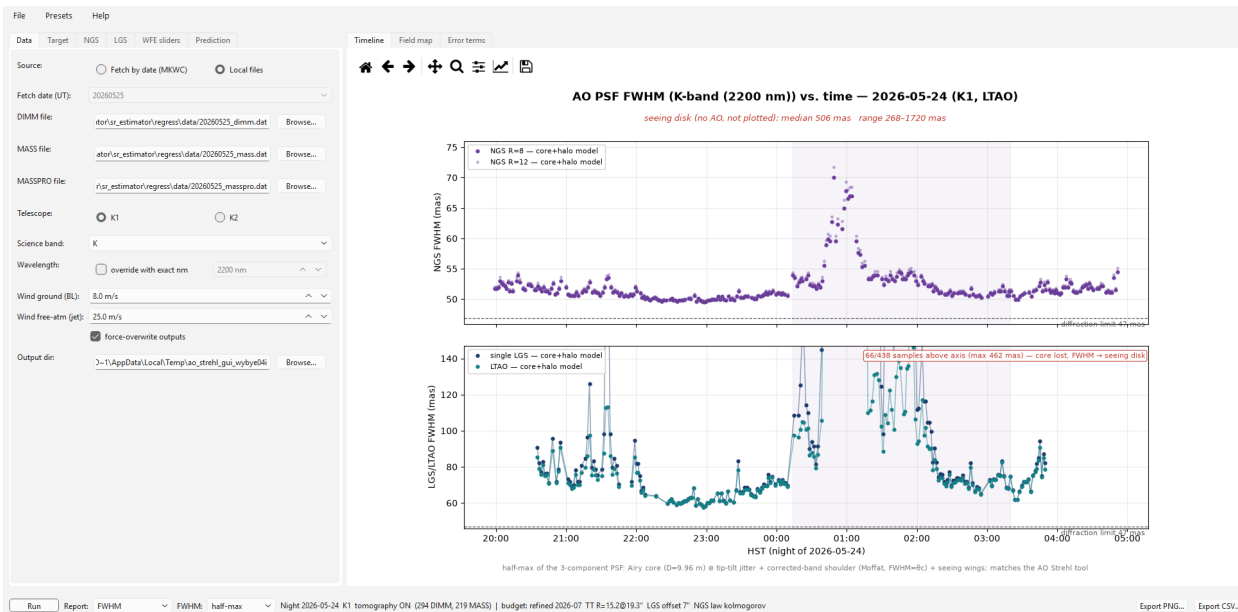


Figure 23 — Report = FWHM on the Timeline, showing the two-panel FWHM view. The FWHM convention combo sits beside the Report selector.

8. Predicting performance for conditions you choose

The Prediction tab answers “what would performance look like if the atmosphere were like this?” It replaces the night's data — on the field map and the Error terms tab — with a synthetic snapshot you dial in, and it needs no night at all: in a fresh GUI with nothing loaded, set your target, sensor and geometry, dial the scenario, and the map and the term breakdown render from those settings alone. This is the true before-the-night workflow — no data download required.

Tick “Use this predicted scenario” to enable it — its tooltip notes that it drives both the field map and the Error terms tab. Set the zenith DIMM and MASS seeing (the sliders run to 2.00” — Mauna Kea seeing is never worse than that, and a saved config with a larger value loads clamped to it), the zenith angle, and either let θ_0 auto-track the synthesized profile (the default) or uncheck “auto” and assert your own. The three presets — Reference (0.5”/0.3”), Free-atm dominated (0.5”/0.45”), and Ground-layer dominated (0.5”/0.15”) — all hold the same total seeing and move turbulence between the ground and the jet, so they show how the *location* of turbulence (not its strength) drives off-axis performance and θ_0 .

Setting the turbulence by layer. The Prediction tab has two pages. *Scenario* is the DIMM/MASS pair described above. *Layers* sets the scenario's turbulence on the LTAO reconstructor's own altitude grid — ground, 0.5, 1, 2, 4, 8 and 16 km above the summit — as turbulence fractions, the reconstructor's own units, which always sum to 1. Tick *Set turbulence by layer* to use it: the total (DIMM) seeing — its row is repeated on this page — sets the scale, and the free-atm (MASS) seeing on the Scenario page is then derived from the aloft share and greyed out, as are the presets. *Reset layers to reconstructor prior* loads the reconstructor's static fractions verbatim ($m = 0$) and is also how you undo edits. Drag a layer and the other six rescale proportionally so the sum stays 1, so turbulence moves between layers rather than appearing or vanishing; to enter an exact set, type the seven values in the *Exact* row and press Apply (they are normalized only if they do not already sum to 1, and the readout says so). The readout gives the derived free-atm and ground seeing, the aloft share, and the layer mismatch m against the prior; the twin-panel Cn^2 profile is repeated on this page so it stays in view while you drag. The field map and Error terms follow every change (Figures 26–29). How the fractions become a profile, and why θ_0 and the anisoplanatism terms move with them, is KAON 1542 §3.8.

While a scenario is active the field-map title is flagged PREDICTED SCENARIO in orange with the full conditions spelled out, and the Conditions selector is disabled. The Error terms tab shows you why — the full budget at your chosen conditions, term by term, single-beacon beside LTAO, with the resulting Strehl for each; if a map looks worse than expected, the bars say which term is doing it. Everything else you have set — WFE sliders, NGS fit, geometry, winds, wavelength — carries into the prediction. The twin-panel plot on the tab shows the synthesized Cn^2 profile the scenario implies. How the profile and the θ_0 re-weighting are built is KAON 1542 §3.8.

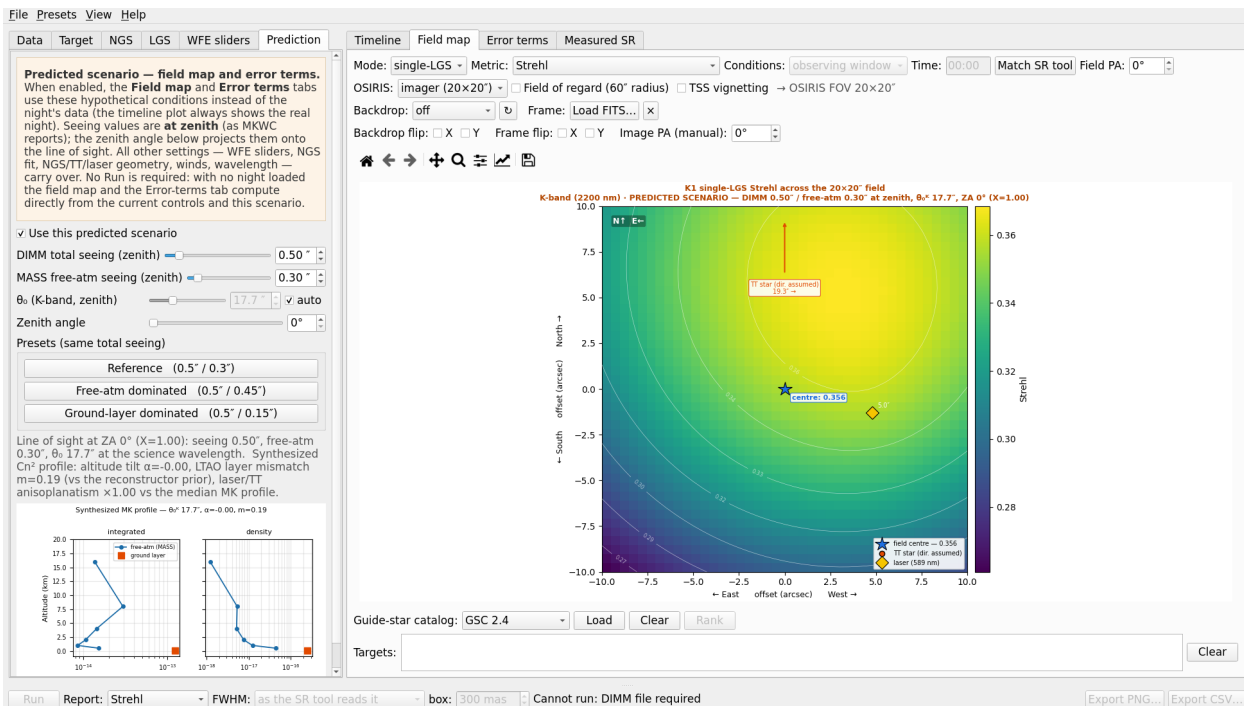


Figure 24 — the Prediction scenario in a fresh GUI with no night loaded: the field map renders from the control settings and the scenario alone (the status bar still reports a Run is impossible), with the PREDICTED SCENARIO title, the seeing / θ_0 / zenith-angle controls and profile presets, and the twin-panel synthesized C_n^2 profile.

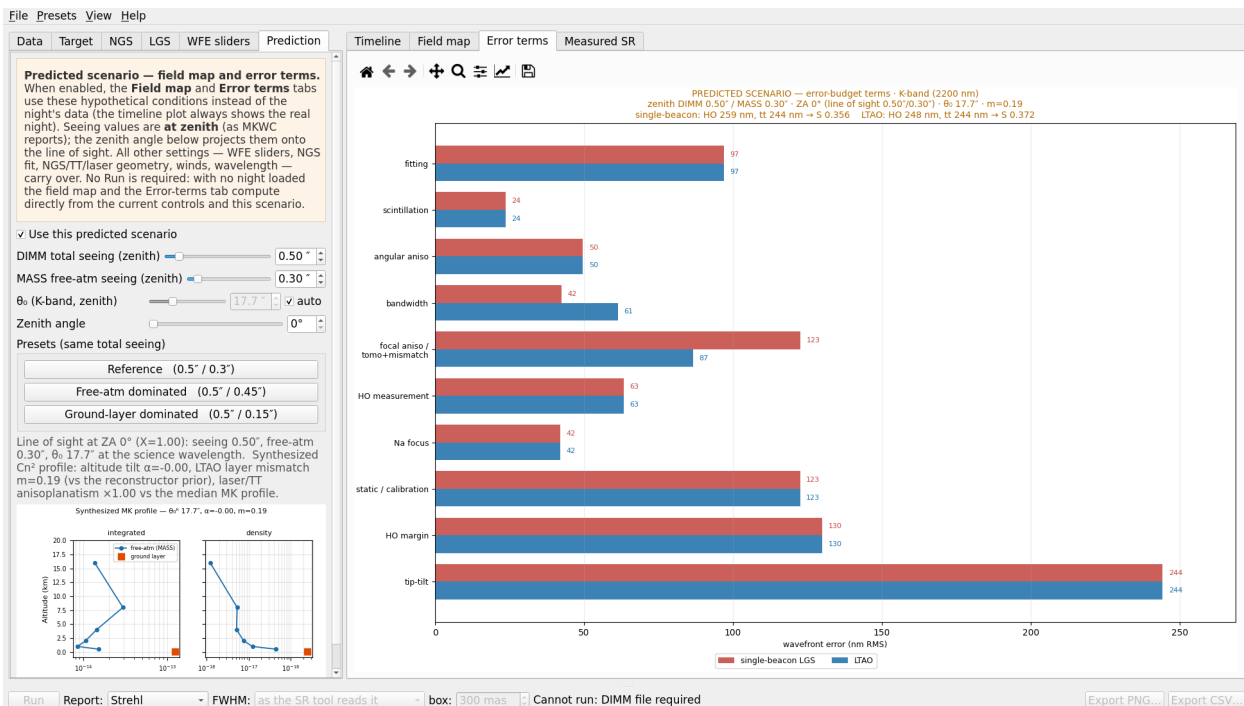


Figure 25 — the Error terms tab under the same scenario: every budget term at the predicted conditions, single-beacon LGS beside LTAO, with the HO/tip-tilt totals and the resulting per-mode Strehl in the title.

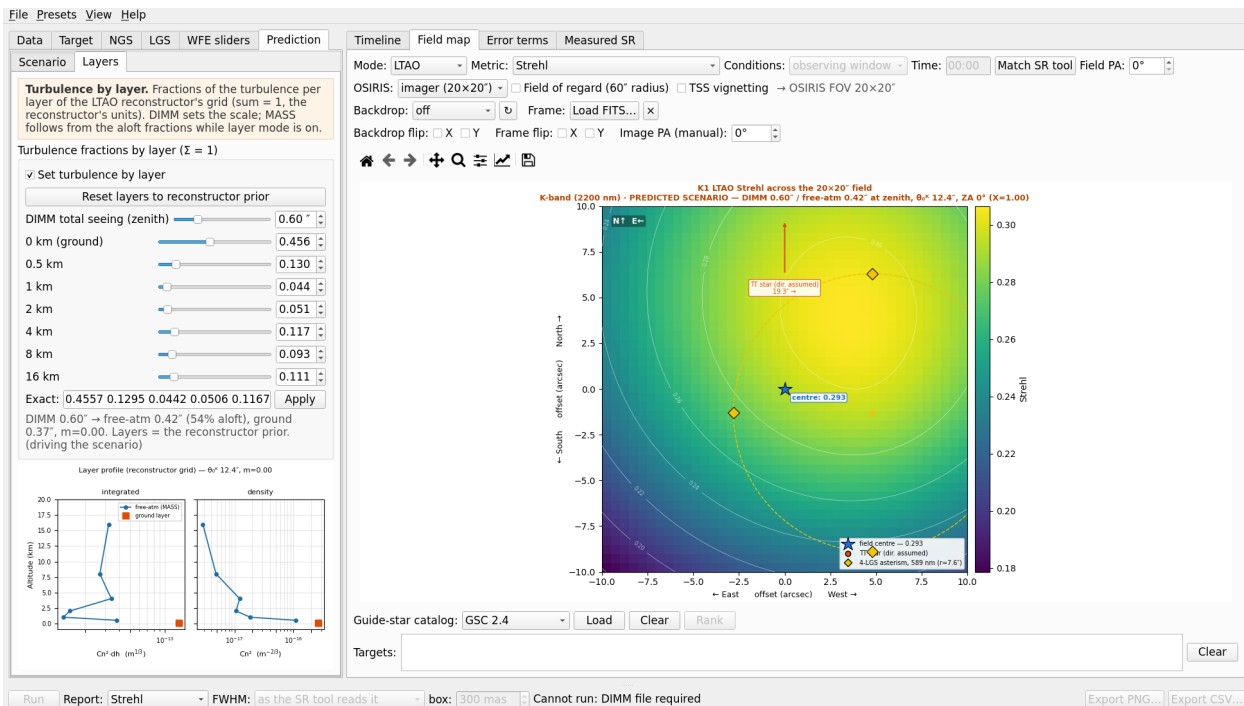


Figure 26 — the Layers page after Reset layers to reconstructor prior, in a fresh GUI with no night loaded: the seven fraction rows at the reconstructor's static values (0.456 / 0.130 / 0.044 / 0.051 / 0.117 / 0.093 / 0.111), the DIMM row at 0.60", the readout deriving free-atm 0.42" (54 % aloft) with $m = 0.00$, the layer-profile plot, and the K1 LTAO Strehl map at those conditions (centre 0.293) with the PREDICTED SCENARIO title spelling out DIMM 0.60" / free-atm 0.42", θ_0^K 12.4".

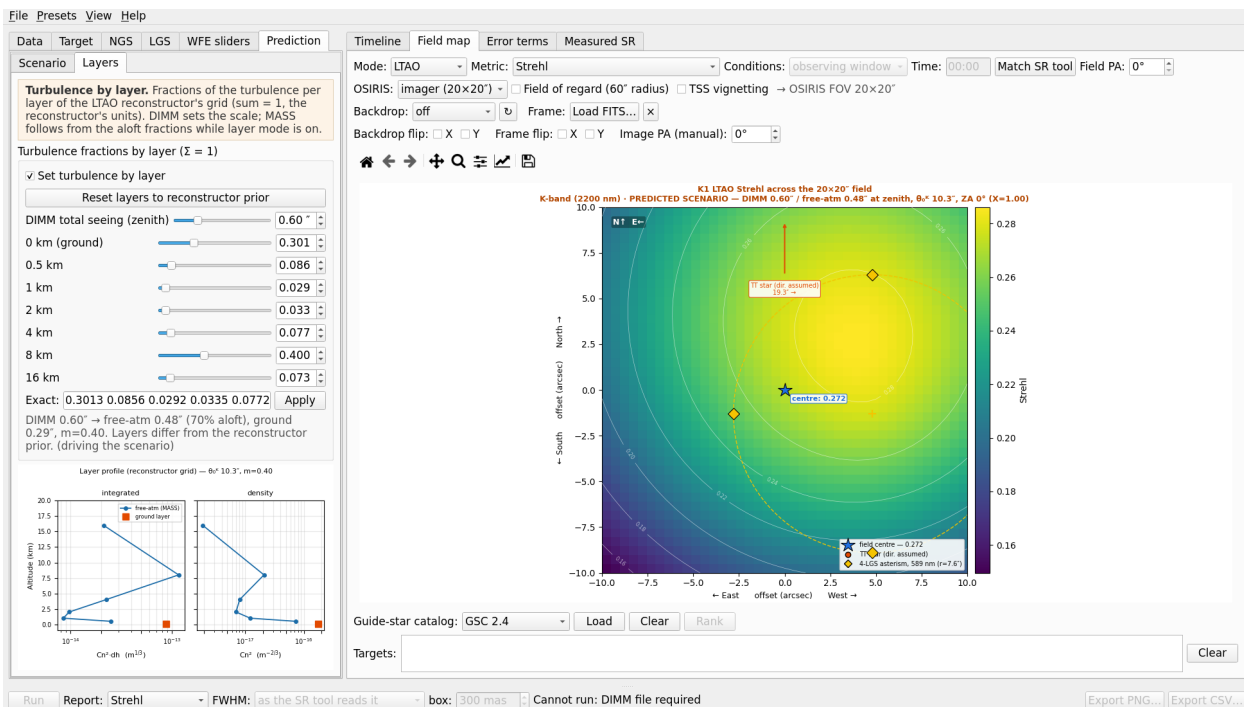


Figure 27 — the same after dragging the 8 km row to 0.40: the other six rows have shrunk proportionally (ground 0.456 → 0.301) so the sum stays 1; the total is unchanged at 0.60" while free-atm rises to 0.48" (70 % aloft), θ_0^K falls to 10.3", m rises to 0.40, the 8 km bin dominates the profile plot, and the LTAO map degrades (centre 0.293 → 0.272).

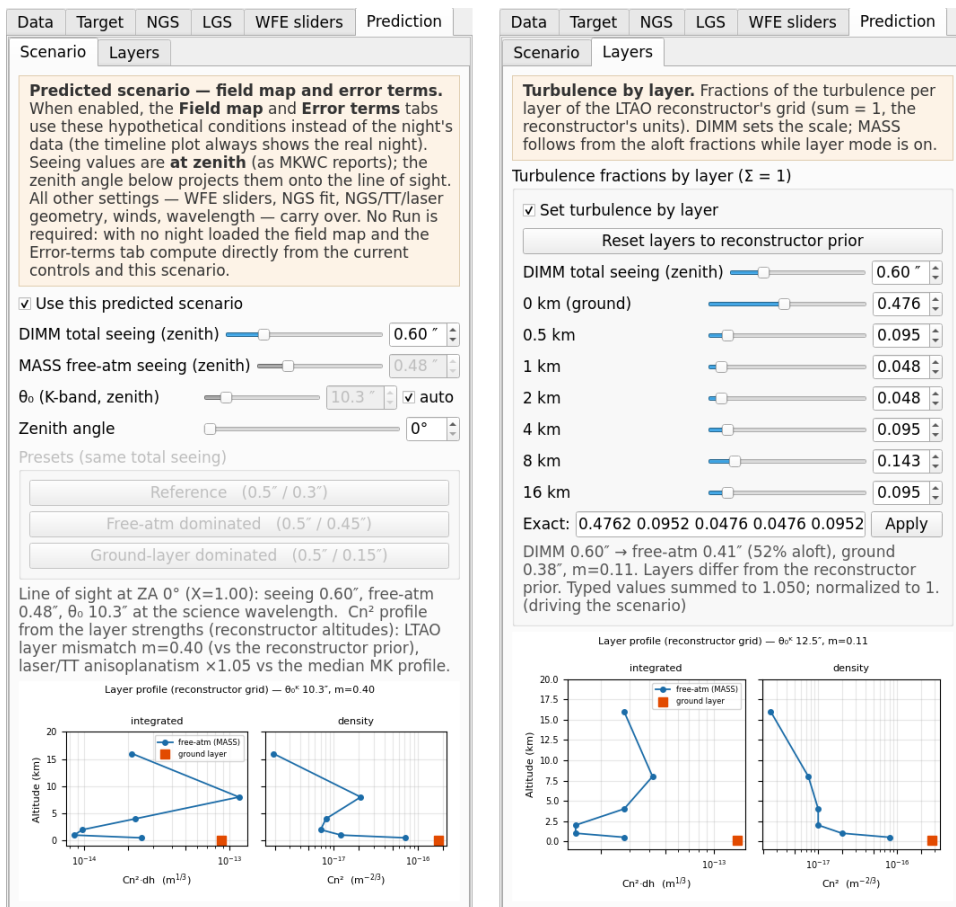


Figure 28 — (a) the Scenario page while layer mode is on: DIMM live, MASS greyed at the derived 0.48", θ_0 auto-tracking the layer profile, the presets disabled, and the readout noting the C_n^2 profile comes from the layer strengths; (b) the Exact row after typing "0.5 0.1 0.05 0.05 0.1 0.15 0.1" and Apply: the values summed to 1.05, so they were normalized (ground 0.476 ...) and the readout says so — the rows are set verbatim otherwise, with no proportional redistribution.

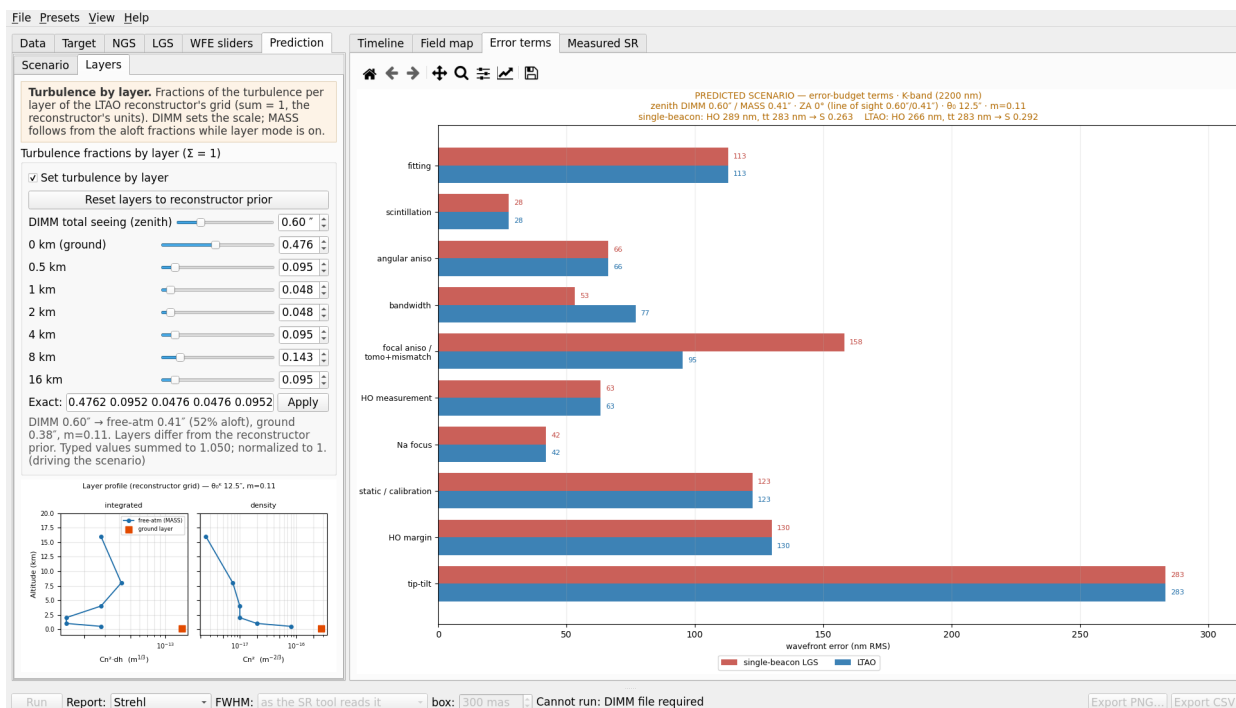


Figure 29 — the Error terms tab under a layer scenario: the title carries the derived pair (DIMM 0.60" / MASS 0.41"), θ_0 12.5" and $m = 0.11$, and the "focal aniso / tomo+mismatch" bar shows the LTAO layer-mismatch penalty at that m (single-beacon 158 nm against LTAO 95 nm).

9. Saving your work

- **Export PNG / Export CSV** (bottom bar, or Ctrl+P / Ctrl+E) write the current figure and the underlying per-sample data; both carry the budget provenance, including any active overrides. On the Field map or Error terms tab, Export PNG saves that tab's own plot rather than the Timeline. The ranking and starlist pop-up tables sort by clicking any column header. Layer settings travel with the config too — layer mode and the seven fractions; configs saved before the Layers page existed load with layer mode off.
- **Save / Load config** (Ctrl+S / Ctrl+O) store every control state as a JSON file, so a scenario can be reopened exactly.
- **Presets menu** offers "Reference budget" (Ctrl+0, clears WFE overrides), "Last night" (fetches yesterday's UT date), and "Tonight's data" (fetches today's UT date, for a night still in progress).

Dark theme. View ► Dark theme restyles the window's controls for low-light use; the plots and everything you export stay in their normal light, print-style rendering either way. Turning on Nighttime mode switches to dark theme automatically (unless you had already chosen it yourself) and switches back when Nighttime mode ends — touch the View menu item yourself at any point and it stops managing the theme for you. Your dark-theme choice is saved with your config; Nighttime mode itself is not.

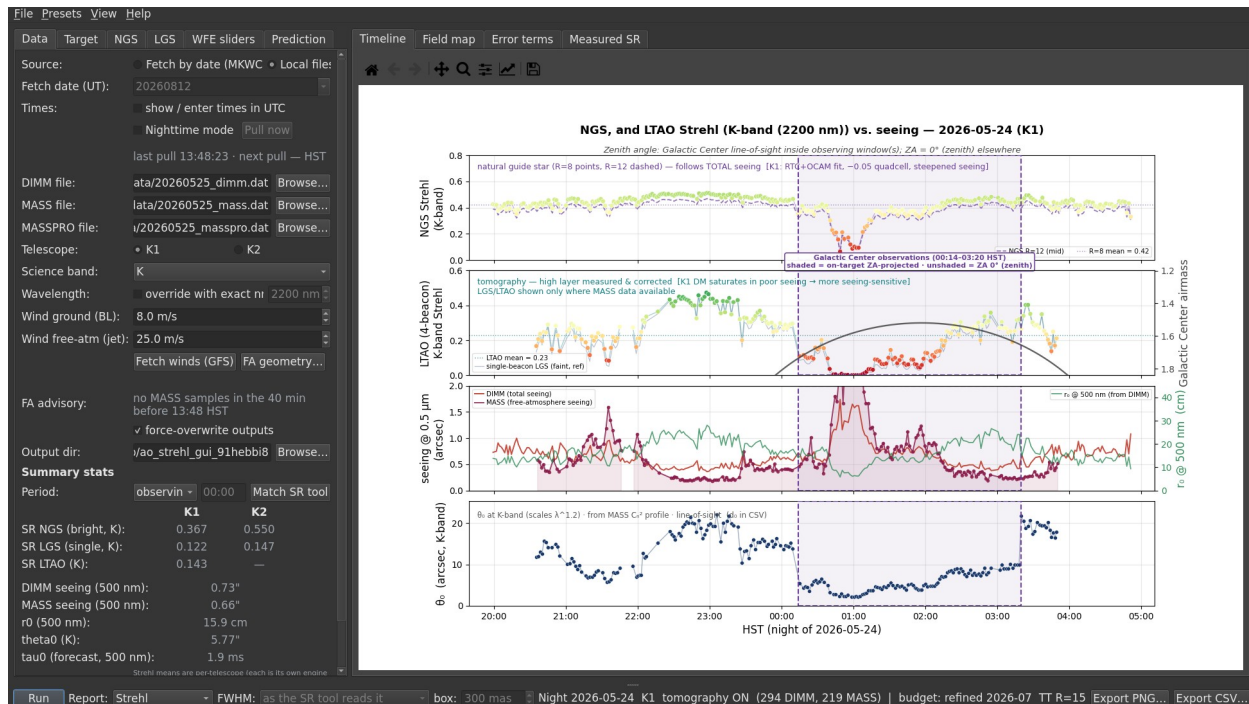


Figure 30 — Dark theme active: the Qt controls restyled dark while the Timeline plot alongside them keeps its normal light, print-style rendering.

Keyboard shortcuts: Ctrl+R Run, Ctrl+S/O save/load config, Ctrl+P/E export PNG/CSV, Ctrl+0 reference budget, Ctrl+Q quit.

10. If a number looks wrong

- Check the status bar first — it names the night, telescope, tomography state and budget, and flags a modified budget. Many surprises are just an override left active or the wrong telescope selected.
- If the field-map centre does not match the timeline, confirm the field-map Conditions snapshot matches the time you have in mind — the map is one instant, the timeline is the whole night.
- **If the field map, the summary stats and the Measured-SR tab disagree on the on-axis Strehl, they are not three models.** All three call the same engine function and agree exactly given the same inputs. Every difference you see is an input difference. The summary stats average the Strehl series over the period while

the field map evaluates a single representative sample, which alone can be worth a couple of hundredths. The Measured-SR tab works at the frame's own wavelength, not the run's. And, most of all, it re-resolves the guide star from the frame itself — so if the frame's real tip-tilt star is much brighter than the magnitude you have configured, its number will be higher, and it is the one that is right. Match SR tool on both panels removes the period and wavelength differences; what is left is the guide star, and that gap is telling you something real. KAON 1542 §10 tabulates all four inputs with their sizes.

- If NGS looks too optimistic or pessimistic at the faint end, check whether the Gompertz fit terms have been edited (NGS tab) — “Reset fit to telescope default” restores them.
- If LTAO looks better than you expected against single-beacon LGS with the tip-tilt star well off axis, check the LTAO TT θ_0 gain (LGS tab). At its default of 1.1 it is a conservative assumption rather than a measurement, and 1.0 turns it off exactly.
- If the Prediction tab's free-atm seeing is not what you typed on the Scenario page, layer mode is on: MASS is then derived from the aloft fractions at the DIMM total ($\epsilon_{fa} = \epsilon_{tot} \times (\text{aloft share})^{(3/5)}$); untick *Set turbulence by layer* to get the pair back.
- **For anything physics-related** — why a term scales as it does, why high-airmass NGS behaves the way it does, what the prediction assumes — the explanation is in KAON 1542. This manual points there rather than reproducing it.